

Conditional and Marginal Inference

In most models the parameter vector can be split into two parts, $\theta = (\psi, \lambda)$, where ψ is the *interest parameter* or *parameter of interest* and λ is a *nuisance* or *incidental parameter*. The former is the focus of enquiry, while the latter summarizes aspects of the model that are not of central concern, but which are nevertheless essential to realistic modelling. Usually ψ has small dimension, often being scalar, while λ may be of high dimension. Different elements of θ may be nuisance or interest parameters at different stages of an investigation. We suppose throughout the discussion below that ψ and λ are variation independent, that is, the parameter space has form $\Psi \times \Lambda$, so knowledge about the range of λ imparts no information about ψ . In many cases it is desirable that inferences be invariant to *interest-preserving reparametrizations*, that is, one-one maps between (ψ, λ) and $(\eta(\psi), \zeta(\psi, \lambda))$.

Example 12.1 (Log-normal mean) If $X \sim N(\mu, \sigma^2)$, then the log-normal variable $Y = \exp(X)$ has mean and variance

$$E(Y) = \exp(\mu + \sigma^2/2) = \psi, \quad \text{var}(Y) = \exp(2\mu + \sigma^2)\{\exp(\sigma^2) - 1\} = \lambda,$$

say. A confidence interval (ψ_-, ψ_+) for ψ should transform to $(\log \psi_-, \log \psi_+)$ if the model is expressed in terms of $\eta = \log \psi$ and $\zeta = \zeta(\psi, \lambda) = \sigma^2$. ■

In many important cases the density of data Y may be factorized as

$$f(y; \psi, \lambda) \propto f(t_1 | a; \psi) f(t_2 | t_1, a; \psi, \lambda),$$

or as

$$f(y; \psi, \lambda) \propto f(t_1 | t_2, a; \psi) f(t_2 | a; \psi, \lambda),$$

where terms free of the parameters have been neglected. The quantity a may not be present, but if it is, it is usually chosen to be an ancillary statistic, a notion discussed in Section 12.1. If such a factorization holds, it is natural to base inference for ψ on the leading term on its right side. Information about ψ can then be extracted from the data without needing to estimate or otherwise account for λ . Leaving aside the presence of a , these terms are respectively marginal and conditional densities of T_1 , and can be

viewed as a *marginal likelihood* and a *conditional likelihood* for ψ . One reason for considering them is that when λ is high-dimensional, the standard likelihood methods described in Chapter 4 and used throughout the book can become unreliable or fail utterly, as in the following classic example.

Example 12.2 (Neyman–Scott problem) The log likelihood based on data y_{ij} , $i = 1, \dots, m$, $j = 1, \dots, k$, supposed to be realized values of independent normal variables with means μ_i and common variance σ^2 , is

So-called because first pointed out by Neyman and Scott (1948).

$$\begin{aligned}\ell(\mu_1, \dots, \mu_m, \sigma^2) &\equiv -\frac{1}{2} \sum_{i=1}^m \left\{ k \log \sigma^2 + \frac{1}{\sigma^2} \sum_{j=1}^k (y_{ij} - \mu_i)^2 \right\} \\ &= -\frac{1}{2} \sum_{i=1}^m \left\{ k \log \sigma^2 + \frac{1}{\sigma^2} \sum_{j=1}^k (y_{ij} - \bar{y}_{i\cdot})^2 + \frac{k}{\sigma^2} (\bar{y}_{i\cdot} - \mu_i)^2 \right\}.\end{aligned}$$

As the maximum likelihood estimate of μ_i is $\bar{y}_{i\cdot} = k^{-1} \sum_j y_{ij}$, the maximum likelihood estimate of σ^2 is $\hat{\sigma}^2 = (mk)^{-1} \sum_{ij} (y_{ij} - \bar{y}_{i\cdot})^2$. The sums of squares $\sum_j (y_{ij} - \bar{y}_{i\cdot})^2$ are independently distributed as $\sigma^2 \chi_{k-1}^2$, so

$$E(\hat{\sigma}^2) = \sigma^2 \frac{(k-1)}{k}, \quad \text{var}(\hat{\sigma}^2) = 2\sigma^4 \frac{(k-1)}{mk^2}.$$

Thus if k is fixed and m increases, $\hat{\sigma}^2$ converges to the wrong value. When $k = 2$, for example, $\hat{\sigma}^2 \xrightarrow{P} \sigma^2/2$, so $\hat{\sigma}^2$ is both biased in small samples and inconsistent as $m \rightarrow \infty$.

The problem here is that the number of parameters increases with the sample size, but the information about each of the μ_i stays fixed. Moreover, even though the expected information for σ^2 does increase with m , it gives a misleading impression of the precision with which σ^2 can be estimated. This is an extreme situation, but similar difficulties arise in many other models. Here they could be eliminated by replacing k by $k - 1$ in the denominator of $\hat{\sigma}^2$, but such a repair is impossible in general. It turns out, however, that use of marginal likelihood rescues the situation. ■

This chapter gives an introduction to conditional and marginal inference. In Section 12.1 we define ancillary statistics and discuss their properties, and in Sections 12.2 and 12.3 describe marginal and conditional likelihoods, which have their major applications for group transformation and exponential family models respectively. In many cases these likelihoods are difficult to construct exactly. Highly accurate approximations to them may be based on saddlepoint procedures, which are closely related to the Bayesian use of Laplace approximation, and these are described in Section 12.3. Finally Section 12.4 gives a brief account of modifications intended to improve inferences based on the profile likelihood.

12.1 Ancillary Statistics

Conditioning arguments have several important roles in statistics. One is to restrict the sample space under consideration, thereby ensuring that repeated sampling

comparisons are made within a so-called *relevant subset* of the full sample space. The conditionality principle (Section 11.1.2) suggests that this will often result in inferences that are close to Bayesian ones, but without the need to specify a prior density that may be difficult to justify. A second role of conditioning is the elimination of nuisance parameters; see Sections 5.2.3, 7.3.3, and 12.3. In this section we focus on the use of conditioning for ensuring relevance of tests and confidence intervals.

Consider a statistical model for data Y that depends on a parameter θ , and suppose that the minimal sufficient statistic for θ is $S = s(Y)$. The factorization theorem implies that inference for θ may be based on the density of S . If we can write $S = (T, A)$, where the distribution of A does not depend on θ , then A is said to be *ancillary for θ* , or, more loosely, an *ancillary statistic*. Note our requirement that A be a function of the minimal sufficient statistic. Some authors do not impose this, and say that any function of Y whose distribution does not depend on θ is ancillary, but we reserve the term *distribution constant* for such statistics; see Section 5.3. We shall see below that in a particular sense ancillary statistics determine the amount of information in the data, and as a result they play a central role in conditional inference.

The likelihood for θ may be written

$$L(\theta) \propto f(s; \theta) = f(t, a; \theta) = f(a)f(t | a; \theta),$$

suggesting that inference for θ be based on the conditional distribution of T given the observed value a of A . One argument for this is that, in principle at least, the experiment that generates S may be regarded as having two stages. The first stage consists of observing a value a from the marginal density of A . In the second stage a value for T is observed from the conditional density of T given that $A = a$. As the distribution of A does not depend upon θ , the conditionality principle implies that inference for θ should be based only on the second stage, which contributes $f(t | a; \theta)$ to the likelihood. According to this argument the subset of the sample space relevant to inference for θ is the set $\{y : a(y) = a_{\text{obs}}\}$, where a_{obs} denotes the value of A actually observed.

Example 12.3 (Uniform density) Let $Y_1, \dots, Y_n$ be a random sample from the density uniform on $(\theta - 1/2, \theta + 1/2)$, with θ unknown. Then the likelihood,

$$L(\theta) = \begin{cases} 1, & \theta - \frac{1}{2} \leq y_1, \dots, y_n \leq \theta + \frac{1}{2}, \\ 0, & \text{otherwise,} \end{cases}$$

can be re-expressed as

$$L(\theta) = \begin{cases} 1, & \max(y_j) - \frac{1}{2} \leq \theta \leq \min(y_j) + \frac{1}{2}, \\ 0, & \text{otherwise,} \end{cases}$$

so the minimal sufficient statistic consists of the smallest and largest order statistics, $S = (Y_{(1)}, Y_{(n)})$. The joint density of $V = Y_{(n)}$ and $U = Y_{(1)}$ is

$$f(u, v) = n(n-1)(v-u)^{n-2}, \quad \theta - \frac{1}{2} \leq u < v < \theta + \frac{1}{2},$$

and on changing variables from (v, u) to (a, t) , where $a = v - u = y_{(n)} - y_{(1)}$ and $t = (y_{(1)} + y_{(n)})/2$, we obtain

$$f(t, a) = n(n-1)a^{n-2}, \quad \theta - \frac{1-a}{2} \leq t \leq \theta + \frac{1-a}{2}, \quad 0 \leq a \leq 1;$$

the random variable $T = (Y_{(1)} + Y_{(n)})/2$ is an unbiased estimator of θ . The support of the joint density for (t, a) is a triangle with vertices $(\theta, 1)$ and $(\theta \pm 1/2, 0)$. Straight-forward calculations give

$$f(t) = n(1-2|t-\theta|)^{n-1}, \quad \theta - \frac{1}{2} \leq t \leq \theta + \frac{1}{2}, \quad (12.1)$$

$$f(a) = n(n-1)(1-a)a^{n-2}, \quad 0 \leq a \leq 1,$$

$$f(t | a; \theta) = \frac{1}{1-a}, \quad \theta - \frac{1-a}{2} \leq t \leq \theta + \frac{1-a}{2}. \quad (12.2)$$

Thus A is ancillary, as is clear from its location-invariance. Conditional on $A = a$, the density of T is uniform on an interval of length $1-a$ centred at θ . From (12.2) we see that the conditional likelihood for θ equals $(1-a)^{-1}$ on the interval with endpoints $t \pm (1-a)/2$ and is zero elsewhere. Thus $a \doteq 0$ conveys little information about θ , whereas $a \doteq 1$ pins down θ very precisely.

For example, a sample of size $n = 2$ with $y_{(1)} = -0.25$ and $y_{(2)} = 0.45$ has $t = 0.1$ and $a = 0.7$. The unconditional equi-tailed 0.9 confidence interval for θ based on (12.1) is $(-0.242, 0.442)$ (Exercise 12.1.2), while the 0.9 conditional interval based on (12.2) is $(-0.035, 0.235)$; their respective lengths are 0.68 and 0.27. The unconditional interval is logically inconsistent with the data, because it includes values such as $\theta = 0.3$ for which $y_{(1)} = -0.25$ could not be observed. The conditional interval takes the observed value of A into account and eliminates such absurdities. ■

Example 12.4 (Regression model) In many regressions the observed explanatory variables x and responses y could both be regarded as realizations of random variables X and Y , in the sense that different values (x, y) might have occurred. This is clearest in an observational study, where individuals each have a number of variables recorded, some being later regarded as explanatory and others as responses. Unless the values of the X s are restricted, the data collection scheme implies that they may be modelled as random variables. If, however, the joint density of (X, Y) factorizes as

$$f(y | x; \psi)f(x),$$

and properties of the marginal distribution of X are not of interest, it will be appropriate to treat X as fixed throughout the analysis, and this is what is generally done in practice. Notice that ψ is a parameter of the conditional density of Y given X ; analysis based on $f(y | x; \psi)$ alone would not be appropriate if ψ also entered the distribution of X .

Consider the linear model $Y = x\beta + \sigma\epsilon$, where Y and x are the vector of responses and the observed design matrix, respectively. The least squares estimator of β is $(x^T x)^{-1} x^T Y$, and conditional on $X = x$, it has variance $\sigma^2(x^T x)^{-1}$. Thus the estimator and its variance both depend on the explanatory variables actually observed, and not on those that might have been observed but in fact were not. For example, if certain

In this example (only) we depart from our convention that X is the observed design matrix.

parameters cannot be estimated from the design matrix actually used, it is irrelevant that with another design they might have been estimable. ■

See Basu (1955, 1958).

Basu's theorem

If S is a complete minimal sufficient statistic, then it is independent of any distribution constant statistic C . To see this in the discrete case, note that for any c and θ the marginal density of C may be written as

$$f_C(c) = \sum_s f_{C|S}(c | s) f_S(s; \theta),$$

where the sum is over all possible values of s . This implies that for all θ

$$\sum_s \{f_C(c) - f_{C|S}(c | s)\} f_S(s; \theta) = 0,$$

and completeness of S gives $f_C(c) = f_{C|S}(c | s)$ for every c and s . Hence C and S are independent. This result is useful, because it assures independence without the effort of computing the joint density of C and S . The argument in the continuous case is analogous.

Example 12.5 (Normal linear model) The minimal sufficient statistic in the normal linear model $y = X\beta + \varepsilon$ consists of $\hat{\beta} = (X^T X)^{-1} X^T y$ and the rescaled sum of squares $S^2 = (y - X\hat{\beta})^T (y - X\hat{\beta}) / (n - p)$. That the pair $(\hat{\beta}, S^2)$ is complete and minimal sufficient follows from properties of the exponential family; see Example 7.10. The standardized residuals

$$\frac{y_j - x_j^T \hat{\beta}}{S(1 - h_j)^{1/2}}, \quad j = 1, \dots, n,$$

have a (degenerate) joint distribution that does not depend on β and σ^2 . Basu's theorem implies that this distribution is independent of $\hat{\beta}$ and S^2 ; see Example 5.14. ■

Location model

Suppose that $y_1, \dots, y_n$ is a random sample from the continuous density $f(y; \theta) = g(y - \theta)$, where $-\infty < \theta < \infty$ and the density g is known. Here θ determines the location of f , and we can write a random variable from f as $Y = \theta + \varepsilon$, where ε has density g . Except for certain special choices of g , such as the normal and uniform densities, the minimal sufficient statistic for θ consists of the order statistics $Y_{(1)} < \dots < Y_{(n)}$, whose joint density is

$$f(y_{(1)}, \dots, y_{(n)}; \theta) = n! \prod_{j=1}^n g(y_{(j)} - \theta), \quad y_{(1)} < \dots < y_{(n)}.$$

We now discuss conditional inference for θ based on the maximum likelihood estimator $\hat{\theta}$. Although rarely of interest in applications, the simplicity of this model brings out the main ideas without unnecessary complication, and helps to motivate later developments.

We assume that $\hat{\theta}$ is the sole solution to the score equation

$$0 = \sum_{j=1}^n \frac{\partial \log g(Y_{(j)} - \hat{\theta})}{\partial \theta} = \sum_{j=1}^n \frac{\partial \log g(A_j)}{\partial \theta}, \quad (12.3)$$

where $A_j = Y_{(j)} - \hat{\theta}$. As $\hat{\theta}$ is equivariant, the random variables $A_1, \dots, A_n$ form a maximal invariant, which is of course distribution constant. Moreover the *configuration* $A = (A_1, \dots, A_n)$ is a function of the minimal sufficient statistic. Hence it is ancillary. The discussion above suggests that inference be based on the conditional distribution of $\hat{\theta}$ given the value a of A . Equivariance of $\hat{\theta}$ means that $Z(\theta) = \hat{\theta} - \theta$ is a pivot, so if we let $z_\alpha(a)$ denote the α quantile of the conditional distribution of $Z(\theta)$, a $(1 - 2\alpha)$ conditional confidence interval for θ will have limits

$$\hat{\theta} - z_{1-\alpha}(a), \quad \hat{\theta} - z_\alpha(a)$$

that depend on the observed a .

Another possibility is to use the unconditional distribution of $Z(\theta)$, whose quantiles z_α and $z_{1-\alpha}$ do not depend on a . These quantiles are readily estimated by simulation, but the $z_\alpha(a)$ are not. In fact the conditional distributions of $\hat{\theta}$ and hence of $Z(\theta)$ are fairly easily found, as we now see.

Owing to the constraint (12.3), the density of A is degenerate, and there exists a function h such that $a_1 = h(a_2, \dots, a_n)$; this is assumed sufficiently smooth for the development below. Thus

$$y_{(1)} = \hat{\theta} + h(a_2, \dots, a_n), \quad y_{(j)} = \hat{\theta} + a_j, \quad j = 2, \dots, n.$$

The Jacobian for transformation from $(y_{(1)}, \dots, y_{(n)})$ to $(\hat{\theta}, a_2, \dots, a_n)$ is

$$\begin{vmatrix} \frac{\partial y_{(1)}}{\partial \theta} & \frac{\partial y_{(2)}}{\partial \theta} & \cdots & \frac{\partial y_{(n)}}{\partial \theta} \\ \frac{\partial y_{(1)}}{\partial a_2} & \frac{\partial y_{(2)}}{\partial a_2} & \cdots & \frac{\partial y_{(n)}}{\partial a_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_{(1)}}{\partial a_n} & \frac{\partial y_{(2)}}{\partial a_n} & \cdots & \frac{\partial y_{(n)}}{\partial a_n} \end{vmatrix} = \begin{vmatrix} 1 & 1 & \cdots & 1 \\ h_1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ h_n & 0 & \cdots & 1 \end{vmatrix} = H(a),$$

say, where $h_j = \partial h(a_2, \dots, a_n) / \partial a_j$. Hence the joint density of $\hat{\theta}$ and A is

$$f(\hat{\theta}, a; \theta) = n! H(a) \prod_{j=1}^n g(y_{(j)} - \theta) \Big|_{y_{(j)} = \hat{\theta} + a_j} = n! H(a) \prod_{j=1}^n g(a_j + \hat{\theta} - \theta),$$

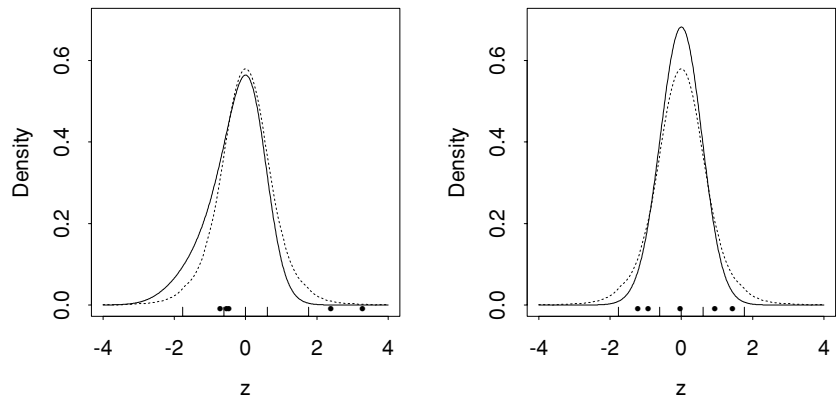
so for $a_1 \leq \dots \leq a_n$ and $\hat{\theta}$ real,

$$f(\hat{\theta} | a; \theta) = \frac{f(\hat{\theta}, a; \theta)}{f(a)} = \frac{\prod_{j=1}^n g(a_j + \hat{\theta} - \theta)}{\int_{-\infty}^{\infty} \prod_{j=1}^n g(a_j + u) du}. \quad (12.4)$$

This conditional density contains all the information in the data concerning θ , appropriately conditioned. Changing variables to $z(\theta) = \hat{\theta} - \theta$ and integrating gives

$$\Pr \{Z(\theta) \leq z \mid A = a\} = \frac{\int_{-\infty}^z \prod_{j=1}^n g(a_j + u) du}{\int_{-\infty}^{\infty} \prod_{j=1}^n g(a_j + u) du}.$$

Figure 12.1 Conditional inference for location in two samples of size $n = 5$ from the t_3 density. Conditional density of pivot $Z(\theta) = \hat{\theta} - \theta$ given configuration ancillary (solid), and unconditional density of $Z(\theta)$ (dotted). The blobs show the observed configuration, and the rug the expected configuration.



This function can be obtained by evaluating the integrand on a grid of values, or by more sophisticated methods. Once it has been approximated the quantiles $z_\alpha(a)$ are easily found.

Example 12.6 (Student t distribution) Figure 12.1 shows the conditional densities of $Z(\theta)$ for two samples of size five from the t_3 density. In the left panel the configuration is very asymmetric, and hence so is the conditional density of $\hat{\theta}$. In the right panel the configuration is quite symmetric but underdispersed relative to a typical configuration. Hence the conditional density of $Z(\theta)$ is more peaked than the unconditional density, thereby giving more precise inferences.

In the left panel the conditional 0.025 and 0.975 quantiles are -2.11 and 1.03 , so if $\hat{\theta} = 0$ the 0.95 confidence interval, $(-1.03, 2.11)$, is asymmetric in the same direction as the configuration. The corresponding confidence interval for the right panel would be $(-1.13, 1.18)$, to be compared to the unconditional interval $(-1.61, 1.61)$. ■

It is suggestive to express $f(\hat{\theta} | a; \theta)$ using the log likelihood for the model, which we write as $\ell(\theta; \hat{\theta}, a)$ to stress its dependence on the data through $\hat{\theta}$ and a . Setting $u = \theta + \hat{\theta} - v$ we see that (12.4) becomes

$$f(\hat{\theta} | a; \theta) = \frac{\exp \{ \ell(\theta; \hat{\theta}, a) \}}{\int_{-\infty}^{\infty} \exp \{ \ell(v; \hat{\theta}, a) \} dv}. \quad (12.5)$$

Laplace approximation to the integral (Section 11.3.1) gives

$$f(\hat{\theta} | a; \theta) = (2\pi)^{-1/2} |J(\hat{\theta}; \hat{\theta}, a)|^{1/2} \exp \{ \ell(\theta; \hat{\theta}, a) - \ell(\hat{\theta}; \hat{\theta}, a) \} \{ 1 + O(n^{-1}) \}. \quad (12.6)$$

where $J(\hat{\theta}; \hat{\theta}, a)$ equals $-\partial^2 \ell(\theta; \hat{\theta}, a) / \partial \theta^2$ evaluated at $\theta = \hat{\theta}$. In fact the quantity $J(\hat{\theta}; \hat{\theta}, a)$ does not depend on $\hat{\theta}$, because the log likelihood may be written as $\ell(\hat{\theta} - \theta)$. By (12.5) we see that renormalizing (12.6) to have unit integral will recover the exact expression, giving

$$f(\hat{\theta} | a; \theta) = (2\pi)^{-1/2} \bar{c}(a) |J(\hat{\theta}; \hat{\theta}, a)|^{1/2} \exp \{ \ell(\theta; \hat{\theta}, a) - \ell(\hat{\theta}; \hat{\theta}, a) \}. \quad (12.7)$$

Detailed analysis of the Laplace expansion shows that $\bar{c}(a) = 1 + O(n^{-1})$.

This is called the p^* formula or Barndorff-Nielsen's formula.

Formula (12.7) expresses the conditional density of $\hat{\theta}$ in terms of the log likelihood and its derivatives and is the basis for much further development. It is exact for group transformation models and approximately true more generally, though we shall prove neither of these assertions; see instead the bibliographic notes.

If we accept that (12.7) is correct, how may it be used for inference? One possibility is to obtain conditional confidence intervals for θ by transforming $\hat{\theta}$ to a normally-distributed pivot. To do this, we write the integral of (12.6) in form (11.30), taking $a(u) = |J(u; u, a)|^{1/2}$ and $g(u) = \ell(u; u, a) - \ell(\theta; u, a)$. Then the argument leading to (11.31) gives

$$\Pr(\hat{\theta} \leq t \mid A = a; \theta) = \Phi \{r^*(\theta)\} + O(n^{-1}), \quad (12.8)$$

where $r^*(\theta) = r(\theta) + r(\theta)^{-1} \log\{v(\theta)/r(\theta)\}$, with

$$r(\theta) = \text{sign}(t - \theta) [2 \{\ell(t; t, a) - \ell(\theta; t, a)\}]^{1/2}, \quad v(\theta) = \frac{\ell_{,\hat{\theta}}(t; t, a) - \ell_{,\hat{\theta}}(\theta; t, a)}{|J(t; t, a)|^{1/2}}.$$

The last of these involves a *sample space derivative* of ℓ ,

$$\ell_{,\hat{\theta}}(\theta; \hat{\theta}, a) = \frac{\partial \ell(\theta; \hat{\theta}, a)}{\partial \hat{\theta}}.$$

In order to obtain this, ℓ must be expressed as a function of $\hat{\theta}$ and a so that it can be differentiated partially with respect to $\hat{\theta}$, holding a fixed. Usually this re-expression is difficult, and approximations to such derivatives are needed. For the location model, however, $\ell_{,\hat{\theta}}(t; t, a) = 0$ for any t , and

$$\ell_{,\hat{\theta}}(\theta; \hat{\theta}, a) = -\frac{\partial \ell(\theta; \hat{\theta}, a)}{\partial \theta} = -\ell_{\theta}(\theta; \hat{\theta}, a),$$

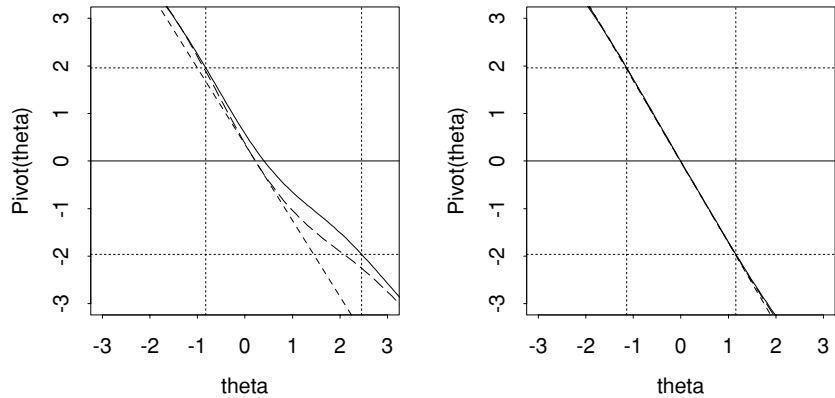
say. Thus $v(\theta) = \ell_{\theta}(\theta; t, a)/|J(t; t, a)|^{1/2}$ is the score statistic.

More detailed computations show that in a *moderate deviation region*, where $n^{1/2}(t - \hat{\theta}) = O(1)$, the error in (12.8) is relative and of order $O(n^{-3/2})$: the right-hand side is in fact $\Phi \{r^*(\theta)\} \{1 + O(n^{-3/2})\}$. In a *large deviation region*, for which $t - \hat{\theta} = O(1)$, the error remains relative but becomes $O(n^{-1})$. The relative error property helps explain why the extraordinary accuracy of this type of approximation persists far into the tails of the distribution of $\hat{\theta}$.

Expression (12.8) shows that the random quantity $R^*(\theta)$ corresponding to $r^*(\theta)$ is an approximate pivot, conditional on $A = a$, because its distribution is almost standard normal. As this distribution is essentially independent of a , $R^*(\theta)$ is also approximately normal unconditionally. The limits of a $(1 - 2\alpha)$ confidence interval for θ may be found as those θ for which $\Phi \{r^*(\theta)\} = \alpha, 1 - \alpha$; equivalently we may solve for θ the equations $r^*(\theta) = \Phi^{-1}(\alpha)$ and $r^*(\theta) = \Phi^{-1}(1 - \alpha)$.

Owing to the structure of the location model, $R^*(\theta)$ is a monotone function of $Z(\theta) = \hat{\theta} - \theta$, so both $Z(\theta)$ and $R^*(\theta)$ yield the same conditional confidence intervals; $R^*(\theta)$ can be regarded as a transformation of $\hat{\theta} - \theta$ to normality. Such transformations have the same effect in much greater generality. There is a close link to the Bayesian approximations of Section 11.3.1.

Figure 12.2 Use of normal approximation to pivotal quantities $r^*(\theta)$ (solid), $r(\theta)$ (large dashes), and $z_1(\theta)$ (small dashes) to construct 0.95 confidence intervals for location based on two samples of size 5. The dotted horizontal lines shows the 0.025 and 0.975 quantiles of the standard normal distribution, and the dotted vertical lines mark the limits of the confidence interval based on $r^*(\theta)$. Note how the confidence intervals based on $r^*(\theta)$ and $r(\theta)$ take account of the skewness of the configuration ancillary.



The approximate pivot $R^*(\theta) = R(\theta) + R_{\text{INF}}(\theta)$, where

$$R_{\text{INF}}(\theta) = R(\theta)^{-1} \log \{V(\theta)/R(\theta)\}, \quad (12.9)$$

is a modified form of the signed likelihood ratio statistic

$$R(\theta) = \text{sign}(\hat{\theta} - \theta) [2 \{ \ell(\hat{\theta}; \hat{\theta}, a) - \ell(\theta; \hat{\theta}, a) \}]^{1/2}.$$

As $R(\theta)$ has an approximate normal distribution with absolute error of $O(n^{-1/2})$ rather than relative error of $O(n^{-3/2})$, $R_{\text{INF}}(\theta)$ can be regarded as a correction to $R(\theta)$ that gives an improved normal approximation; although $R^*(\theta)$ is slightly more complicated, it yields inferences that are much more accurate. It can be useful to compute values $r_{\text{INF}}(\theta)$ of $R_{\text{INF}}(\theta)$ for the plausible range of values of θ .

Example 12.7 (Student t distribution) The log likelihood contribution for the t_ν density is $g(u) = -\frac{1}{2}(\nu + 1) \log(1 + u^2/\nu)$, so the log likelihood and its derivatives are

$$\ell(\theta; \hat{\theta}, a) = -\frac{\nu + 1}{2} \sum_{j=1}^n \log \{1 + (a_j + \hat{\theta} - \theta)^2/\nu\},$$

$$\ell_{\theta;}(\theta; t, a) = (\nu + 1) \sum_{j=1}^n (a_j + t - \theta) / \{ \nu + (a_j + t - \theta)^2 \},$$

$$J(t; t, a) = (\nu + 1) \sum_{j=1}^n (\nu - a_j^2) / (\nu + a_j^2)^2,$$

from which $r(\theta)$, $v(\theta)$, and $r^*(\theta)$ may be found.

The data used in the left panel of Figure 12.1 have $n = 5$, $\hat{\theta} = 0.22$, and configuration $a = (-0.71, -0.54, -0.47, 2.39, 3.28)$. The left panel of Figure 12.2 compares the values of $R^*(\theta)$, $R(\theta)$ and the approximate pivot $Z_1(\theta) = J(\hat{\theta}; \hat{\theta}, a)^{1/2}(\hat{\theta} - \theta)$ derived from the large-sample $N(\theta, J(\hat{\theta}; \hat{\theta}, a)^{-1})$ distribution of $\hat{\theta}$. All three decrease as functions of θ , but $r^*(\theta)$ and $r(\theta)$ are curved for positive θ and shifted upwards relative to $z_1(\theta)$, reflecting the asymmetry of the configuration. The limits of confidence

intervals based on $R^*(\theta)$ are read off as those values for which $r^*(\theta) = z_\alpha$, $z_{1-\alpha}$, and likewise with $r(\theta)$ and $z_1(\theta)$. Thus 0.95 confidence intervals based on $r^*(\theta)$, $r(\theta)$, and $z_1(\theta)$ are $(-0.83, 2.45)$, $(-0.87, 2.08)$, and $(-1.00, 1.44)$. These intervals reflect the upward shift and curvature in $r^*(\theta)$, as the corresponding interval is moved to the right and quite asymmetric.

The right panel of Figure 12.2 shows that for the symmetric configuration in the right of Figure 12.1, confidence intervals based on $Z(\theta)$, $R(\theta)$, and $R^*(\theta)$ are essentially identical. ■

Applications are usually much more complicated than this, and though it nicely illustrates how conditioning can affect inference, one might wonder how relevant the above toy example is to a real world in which models contain nuisance parameters and have no exact ancillary statistics. In practice a key aspect is not the number of observations, but the information they contain. If ten parameters are estimated from 50 observations, say, then one may worry that inference for one key parameter is weakened by having to deal with the other nine: the proportion of parameters to observations is the same as in the example above. Then small-sample procedures may in some cases provide reassurance that standard inferences are adequate, while giving improvements when they are not.

Difficulties with ancillaries

One difficulty in a general treatment of conditional inference is that some statistical models admit no exact ancillary statistic, while others have several. An instance of this is the location model, where any subset of the configuration is ancillary, but there it is clear that the sample space is partitioned most finely by conditioning on the entire configuration, which is a maximal ancillary statistic. Such arguments may not always be applied, however.

Example 12.8 (Multinomial distribution) Consider the multinomial distribution with denominator m and responses $R_1, \dots, R_4$ corresponding to cells with probabilities

$$\pi_1 = \frac{1}{6}(1 + \theta), \quad \pi_2 = \frac{1}{6}(1 - \theta), \quad \pi_3 = \frac{1}{6}(2 + \theta), \quad \pi_4 = \frac{1}{6}(2 - \theta), \quad -1 < \theta < 1.$$

The minimal sufficient statistic is $S = (R_1, R_2, R_3)$, but $A_1 = R_1 + R_2$ and $A_2 = R_2 + R_3$ are binomial with success probabilities $\frac{1}{3}$ and $\frac{1}{2}$. As A_1 and A_2 are functions of S whose distributions are free of θ , both are ancillary. They are not jointly ancillary, however (Problem 12.6).

Conditioning on $A_1 = a_1$ splits the multinomial likelihood into two binomial contributions, one from R_1 with denominator a_1 and probability $\frac{1}{2}(1 + \theta)$, and the other from R_3 with denominator $m - a_1$ and probability $\frac{1}{4}(2 + \theta)$, so the expected information is the sum of the information from the two binomials, that is $I_1(\theta) = 4a_1/(1 - \theta^2) + 16(m - a_1)/(2 - \theta^2)$. Conditioning on A_2 gives a similar

split, but the conditional expected information is

$$I_2(\theta) = 9a_2/\{(1 - \theta)(2 + \theta)\} + 9(m - a_2)/\{(1 + \theta)(2 - \theta)\}.$$

Evidently there is no reason that $I_1(\theta)$ should equal $I_2(\theta)$, and therefore conditional large-sample confidence intervals for θ would depend on whether A_1 or A_2 was used. ■

This lack of uniqueness is annoying in principle but turns out not to be crucial in applications. More awkward difficulties arise when there is no exact ancillary. Then it is necessary to construct approximate ancillaries, and several approaches to this have been proposed. One conceptually simple notion can be used when the model under study can be regarded as nested within a density $f(y; \theta, \lambda)$, with $\lambda = \lambda_0$ corresponding to the model under study. Suppose that the likelihood for the larger model may be expressed as a function of the data through $(\hat{\theta}, \hat{\lambda}, a_0)$, where $\hat{\theta}$ and $\hat{\lambda}$ are maximum likelihood estimators and a_0 is ancillary for (θ, λ) . Let $\hat{\theta}_0$ be the maximum likelihood estimator of θ when $\lambda = \lambda_0$. Then if the model under study is correct, the likelihood ratio statistic $2\{\ell(\hat{\theta}, \hat{\lambda}; a_0) - \ell(\hat{\theta}_0, \lambda_0; a_0)\}$ has an approximate chi-squared distribution whatever the value of θ and so is approximately ancillary. More elaborate variants of this argument are required to produce ancillary statistics on which conditioning can readily be performed in practice, for example by decomposing the ancillary into a sum of squared approximate standard normal variables, and conditioning on these. It may also be necessary to modify these variables to have a joint normal distribution to higher order. The details are tedious and since explicit forms of these ancillaries are not needed below we shall not delve into them. Although there are different forms of approximate ancillary, most of the approximations in later sections do not depend on the particular one used.

Exercises 12.1

- 1 Let $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu, c^2\mu^2)$, with c known. Show that $\bar{Y}/S$ is ancillary for μ .
- 2 In Example 12.3, show that $(1 - 2\alpha)$ equi-tailed conditional and unconditional confidence intervals are $t \pm (1/2 - \alpha)(1 - a)$ and $t \pm \{(2\alpha)^{1/n} - 1\}/2$. Hence verify the intervals given in the example.
- 3 If $Y_1, \dots, Y_n$ is an exponential random sample, show that the joint distribution of $Y_1/S, \dots, Y_n/S$ is independent of that of $S = \sum Y_j$, without computing it. Is this true when S is replaced by any other equivariant estimator of scale?
- 4 Let $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ with σ^2 known. Show that $(Y_1 - \bar{Y}, \dots, Y_n - \bar{Y})$ is distribution constant, and deduce that $\bar{Y}$ and $\sum (Y_j - \bar{Y})^2$ are independent.
- 5 When is the configuration $(Y_1 - \hat{\eta})/\hat{\tau}, \dots, (Y_n - \hat{\eta})/\hat{\tau}$ in a location-scale model ancillary as well as maximal invariant?
- 6 Show that conditional and unconditional inference for the mean of the normal location-scale model are the same.
- 7 Give expressions for the quantities needed to compute $r^*(\theta)$ for the location model when the data have the Gumbel density, so $g(u) = -u - e^{-u}$. Hence find confidence intervals based on $r(\theta)$, $r^*(\theta)$, and $z_1(\theta)$ with the data $-0.32, -0.49, -0.25, 2.61, 3.50$ from Example 12.6.

12.2 Marginal Inference

Consider a model for data with minimal sufficient statistic (T_1, T_2, A) , where A is ancillary, and for which

$$f(y; \psi, \lambda) \propto f(t_1, t_2, a; \psi, \lambda) = f(a)f(t_1 | a; \psi)f(t_2 | t_1, a; \psi, \lambda).$$

As the density of T_1 conditional on A depends only on ψ , inference for ψ may be based on the *marginal likelihood* $L_m(\psi) = f(t_1 | a; \psi)$. In many cases there is no ancillary, and then $L_m(\psi)$ is simply the marginal density of T_1 . There may be some loss of information about ψ due to neglecting the third term in the decomposition above, but the complications in retrieving it are presumed to outweigh the benefits.

We have already encountered marginal likelihood and maximum marginal likelihood estimators in several contexts. Such estimators and derived statistics will have the usual large-sample distributions provided that the usual regularity conditions apply to $f(t_1 | a; \psi)$.

Example 12.9 (Normal linear model) Consider the usual normal linear model $y = X\beta + \sigma\varepsilon$, where the ε_j are independent standard normal variables and β has dimension $p \times 1$. The minimal sufficient statistics for (β, σ^2) are

$$T_1 = S^2 = (n - p)^{-1}(y - X\hat{\beta})^\top(y - X\hat{\beta}), \quad T_2 = \hat{\beta} = (X^\top X)^{-1}X^\top y,$$

and these are independent, with $(n - p)S^2/\sigma^2$ having a χ^2_{n-p} distribution. If the interest parameter is σ^2 , and β is the nuisance parameter, we have

$$f(y; \sigma^2, \beta) = f(y | \hat{\beta}, s^2)f(\hat{\beta}; \beta, \sigma^2)f(s^2; \sigma^2).$$

As the density of S^2 depends only on σ^2 , we may base inference for σ^2 on the corresponding marginal likelihood

$$L_m(\sigma^2) = f(s^2; \sigma^2) = \left(\frac{n - p}{2\sigma^2}\right)^{(n-p)/2} \frac{(s^2)^{(n-p)/2-1}}{\Gamma\left(\frac{n-p}{2}\right)} \exp\left\{-\frac{(n - p)s^2}{2\sigma^2}\right\},$$

where $\sigma^2 > 0$, $s^2 > 0$. The marginal maximum likelihood estimate of σ^2 is s^2 , with expected information $(n - p)/(2\sigma^4)$ from the marginal likelihood. These give inferences for σ^2 essentially equivalent to those described in Chapter 8.

This argument applies to any normal linear model, and in particular to Example 12.2, where $p = k$ and

$$\begin{aligned} S^2 &= \frac{1}{n - p}(y - X\hat{\beta})^\top(y - X\hat{\beta}) \\ &= \frac{1}{m(k - 1)} \sum_{i=1}^m \sum_{j=1}^k (y_{ij} - \bar{y}_i)^2 \sim \frac{\sigma^2}{m(k - 1)} \chi^2_{m(k-1)}. \end{aligned}$$

Here marginal likelihood automatically produces corrected inferences for σ^2 . ■

Example 12.10 (Partial likelihood) The partial likelihood used with the proportional hazards model (Section 10.8.2) is a marginal likelihood. Suppose that survival

time Y has hazard $\xi h(y)$, so its density may be written $\xi h(y) \exp\{-\xi H(y)\}$, where $H(y) = \int_0^y h(u) du$ is the baseline cumulative hazard. Observe that

$$\int_u^\infty h(s) e^{-\gamma H(s)} ds = \gamma^{-1} e^{-\gamma H(u)}.$$

To illustrate the argument, consider $n = 4$ continuous observations that fall in the order $0 < Y_2 < Y_3^+ < Y_1 < Y_4$, where Y_3^+ is right-censored. With nothing known about the baseline hazard $h(y)$, a minimal sufficient statistic is the set of failure times and censoring indicators $(Y_1, 1), (Y_2, 1), (Y_3, 0), (Y_4, 1)$; these are in 1–1 correspondence with the order statistics $(Y_{(1)}, Y_{(2)}, Y_{(3)}, Y_{(4)})$ and inverse ranks $(2, 3^+, 1, 4)$, the j th of the inverse ranks giving the original index of $Y_{(j)}$ and its censoring status. We shall compute the probability of seeing this particular realization of inverse ranks. If Y_3 had been observed, the joint density of the Y_j would be

$$\xi_2 h(y_2) e^{-\xi_2 H(y_2)} \times \xi_3 h(y_3) e^{-\xi_3 H(y_3)} \times \xi_1 h(y_1) e^{-\xi_1 H(y_1)} \times \xi_4 h(y_4) e^{-\xi_4 H(y_4)},$$

so the probability that $0 < Y_2 < Y_1 < Y_4$ with Y_3 censored somewhere to the right of Y_2 is

$$\xi_2 h(y_2) e^{-\xi_2 H(y_2)} \times e^{-\xi_3 H(y_2)} \times \xi_1 h(y_1) e^{-\xi_1 H(y_1)} \times \xi_4 h(y_4) e^{-\xi_4 H(y_4)}.$$

Hence the probability that the uncensored observations fail in the order observed, with Y_3 censored to the right of Y_2 , is

$$\xi_2 \xi_1 \xi_4 \int_0^\infty dy_2 \int_{y_2}^\infty dy_1 \int_{y_1}^\infty dy_4 h(y_2) h(y_1) h(y_4) e^{-(\xi_2 + \xi_3) H(y_2) - \xi_1 H(y_1) - \xi_4 H(y_4)},$$

and this equals

$$\frac{\xi_2}{\xi_1 + \xi_2 + \xi_3 + \xi_4} \times \frac{\xi_1}{\xi_1 + \xi_4} = \prod_j \frac{\xi_j}{\sum_{i \in \mathcal{R}_j} \xi_i},$$

where the product is over those j for which Y_j is uncensored and $\mathcal{R}_j$ denotes the risk set of individuals available to fail at the j th failure time. This last expression is simply the partial likelihood (10.58) in this case. Plainly this argument generalizes to n failures, the only complication being notational. Thus partial likelihood is a marginal likelihood based on the inverse ranks of the failure and censoring times. ■

Restricted maximum likelihood

An important application of marginal likelihood is to normal mixed models, which we met in Section 9.4.2. There the response vector may be written

$$y = X\beta + Zb + \varepsilon, \quad (12.10)$$

where the $n \times p$ and $n \times q$ matrices X and Z are known, β is an unknown $p \times 1$ parameter vector, and the $q \times 1$ and $n \times 1$ random vectors b and ε are independent with respective $N_q(0, \Omega_b)$ and $N_n(0, \sigma^2 I_n)$ distributions. Suppose that the variance matrix $\sigma^2 \Upsilon^{-1} = \sigma^2 I_n + Z\Omega_b Z^T$ exists and that Υ depends on parameters ψ but not on β . We aim to construct a marginal likelihood for σ^2 and ψ , eliminating the nuisance parameter β .

We can write $y = X\beta + \zeta$, where $\zeta \sim N_n(0, \sigma^2\Upsilon^{-1})$. If ψ and hence Υ were known, the maximum likelihood estimator of β would be

$$\widehat{\beta}_\psi = (X^T\Upsilon X)^{-1}X^T\Upsilon y = \beta + (X^T\Upsilon X)^{-1}X^T\Upsilon\zeta,$$

whose distribution is $N_p(\beta, \sigma^2(X^T\Upsilon X)^{-1})$.

Let H denote the $n \times n$ matrix $X(X^T X)^{-1}X^T$; it satisfies $HX = X$. A natural basis for building a marginal likelihood is the vector of residuals $(I_n - H)y$, whose distribution does not depend on β , but as $I_n - H$ has rank $(n - p)$ this distribution is degenerate and it seems better to take just $(n - p)$ linearly independent residuals. Consider therefore the $(n - p) \times 1$ random variable $U = B^T y$, where B is any $n \times (n - p)$ matrix with $BB^T = I_n - H$ and $B^T B = I_{n-p}$ (Exercise 12.2.3); B has rank $(n - p)$. Now $B^T y = B^T B B^T y = B^T(I_n - H)y$ is a linear combination of the residuals. Furthermore

$$B^T X = B^T B B^T X = B^T(I_n - H)X = 0,$$

giving

$$U = B^T y = B^T(X\beta + \zeta) = B^T\zeta,$$

whose distribution does not depend on β . Moreover U and $\widehat{\beta}_\psi$ are normal variables, with covariance

$$\begin{aligned} E\{U(\widehat{\beta}_\psi - \beta)\} &= B^T E(\zeta\zeta^T)\Upsilon X(X^T\Upsilon X)^{-1} \\ &= \sigma^2 B^T\Upsilon^{-1}\Upsilon X(X^T\Upsilon X)^{-1} = 0, \end{aligned}$$

because $B^T X = 0$. Hence U and $\widehat{\beta}_\psi$ are independent, and therefore

$$f(u; \psi) = \frac{f(u; \psi)f(\widehat{\beta}_\psi; \beta, \psi)}{f(\widehat{\beta}_\psi; \beta, \psi)} = \frac{f(u, \widehat{\beta}_\psi; \beta, \psi)}{f(\widehat{\beta}_\psi; \beta, \psi)} = \frac{f(y; \beta, \psi)}{f(\widehat{\beta}_\psi; \beta, \psi)} \left| \frac{\partial y}{\partial(u, \widehat{\beta}_\psi)} \right|, \quad (12.11)$$

while the Jacobian for the change of variable from $(u, \widehat{\beta}_\psi)$ to y is

$$\begin{aligned} \left| \frac{\partial(u, \widehat{\beta}_\psi)}{\partial y} \right| &= |B \quad X(X^T X)^{-1}| \\ &= \left| \begin{pmatrix} B^T \\ (X^T X)^{-1}X^T \end{pmatrix} (B \quad X(X^T X)^{-1}) \right|^{1/2} \\ &= \left| \begin{pmatrix} B^T B & B^T X(X^T X)^{-1} \\ (X^T X)^{-1}X^T B & (X^T X)^{-1} \end{pmatrix} \right|^{1/2} \\ &= \left| \begin{pmatrix} I_{n-p} & 0 \\ 0 & (X^T X)^{-1} \end{pmatrix} \right|^{1/2} = |X^T X|^{-1/2}. \end{aligned}$$

On substituting this and the normal densities of y and $\widehat{\beta}_\psi$ into (12.11), we find

$$f(u; \psi) = \frac{|X^T X|^{1/2}|\Upsilon|^{1/2}}{(2\pi\sigma^2)^{(n-p)/2}|X^T\Upsilon X|^{1/2}} \exp \left\{ -\frac{1}{2\sigma^2}(y - X\widehat{\beta}_\psi)^T\Upsilon(y - X\widehat{\beta}_\psi) \right\}, \quad (12.12)$$

in which B does not appear. Hence (12.12) is independent of the choice of B and therefore of the linear combination of elements of $(I_n - H)y$ used.

Expression (12.12) is the marginal likelihood on which inference for ψ is based. It is also known as the *restricted likelihood*, because its parameter space involves σ^2 and ψ alone. The log restricted likelihood is

$$\begin{aligned} \ell_m(\psi, \sigma^2) \equiv & \frac{1}{2} \log |\Upsilon| - \frac{1}{2} \log |X^T \Upsilon X| - \frac{1}{2\sigma^2} (y - X\hat{\beta}_\psi)^T \Upsilon (y - X\hat{\beta}_\psi) \\ & - \frac{n-p}{2} \log \sigma^2, \end{aligned} \quad (12.13)$$

where Υ and $\hat{\beta}_\psi$ depend on ψ . This can be maximized with respect to ψ by a Newton–Raphson procedure, leading to a *restricted iterative generalized least squares* algorithm, or using the slower but more stable EM algorithm. Computational considerations are often important in practice, because the matrices involved can be very large.

The profile log likelihood for ψ and σ^2 based directly on the density of y is

$$\ell_p(\psi, \sigma^2) \equiv \frac{1}{2} \log |\Upsilon| - \frac{1}{2\sigma^2} (y - X\hat{\beta}_\psi)^T \Upsilon (y - X\hat{\beta}_\psi) - \frac{n}{2} \log \sigma^2,$$

from which $\ell_m(\psi)$ differs by the addition of the term $\frac{p}{2} \log \sigma^2 - \frac{1}{2} \log |X^T \Upsilon X|$, of order p . Thus if the dimension of β is large, the residual maximum likelihood estimator obtained from $\ell_m(\psi)$ may differ substantially from the usual maximum likelihood estimator from $\ell_p(\psi)$. When the term Zb does not appear in (12.10), $\Upsilon = I_n$ and the estimator of σ^2 obtained by maximizing (12.13) is the usual unbiased quantity (Exercise 12.2.4). Thus the argument leading to (12.13) generalizes that in Example 12.9.

We now illustrate how restricted likelihood can lead to modified inferences; see also Example 9.18.

Example 12.11 (Short time series) Consider m independent individuals on each of which k measurements y_{ij} are available from the distribution

$$\begin{pmatrix} Y_{i1} \\ \vdots \\ Y_{ik} \end{pmatrix} \sim N_k \left\{ \begin{pmatrix} \mu_i \\ \vdots \\ \mu_i \end{pmatrix}, \frac{\sigma^2}{1-\rho^2} \begin{pmatrix} 1 & \rho & \cdots & \rho^{k-1} \\ \rho & 1 & \cdots & \rho^{k-2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho^{k-1} & \rho^{k-2} & \cdots & 1 \end{pmatrix} \right\}, \quad i = 1, \dots, m.$$

Denote the matrix here by Ω_0 . Its determinant is $(1-\rho^2)^{k-1}$ and its inverse has tridiagonal form

$$\frac{1}{1-\rho^2} \begin{pmatrix} 1 & -\rho & 0 & \cdots & 0 & 0 \\ -\rho & 1+\rho^2 & -\rho & \ddots & & 0 \\ 0 & -\rho & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & -\rho & 0 \\ 0 & & \ddots & -\rho & 1+\rho^2 & -\rho \\ 0 & 0 & \cdots & 0 & -\rho & 1 \end{pmatrix}.$$

It is also called the residual likelihood. Restricted maximum likelihood estimation is often abbreviated to REML.

Or RIGLS algorithm.

This formulation is typical of that arising in longitudinal data, where individuals are measured at successive time points. Correlation among measurements on a single individual is represented by Ω_0 , here corresponding to a first-order Markov chain; see Example 6.13. To illustrate the argument above, we construct the profile and marginal log likelihoods for the correlation parameter ρ , with $m + 1$ nuisance parameters $\mu_1, \dots, \mu_m, \sigma^2$.

The matrices X and $\sigma^2 \Upsilon^{-1}$ are of side $mk \times m$ and $mk \times mk$ respectively and

$$X = \begin{pmatrix} 1_k & 0 & \cdots & 0 \\ 0 & 1_k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1_k \end{pmatrix}, \quad \sigma^2 \Upsilon^{-1} = \sigma^2 (1 - \rho^2)^{-1} \text{diag} \{ \Omega_0, \dots, \Omega_0 \},$$

and some algebra gives

$$\frac{1}{2} \log |\Upsilon| = \frac{m}{2} \log(1 - \rho^2), \quad -\frac{1}{2} \log |X^T \Upsilon X| = -\frac{m}{2} \log [(1 - \rho) \{k - (k - 2)\rho\}].$$

More algebra establishes that $\hat{\mu}_i = \bar{y}_i$, so

$$(y - X\hat{\beta}_\psi)^T \Upsilon (y - X\hat{\beta}_\psi) = \sum_{i=1}^m T_i(\bar{y}_i, \rho),$$

where

$$T_i(\mu, \rho) = \sum_{j=1}^k (y_{ij} - \mu)^2 + \rho^2 \sum_{j=2}^{k-1} (y_{ij} - \mu)^2 - 2\rho \sum_{j=1}^{k-1} (y_{ij} - \mu)(y_{i,j+1} - \mu).$$

Thus the profile log likelihood for ρ is given by

$$\ell_p(\rho) = \frac{m}{2} \log(1 - \rho^2) - \frac{mk}{2} \log \hat{\sigma}_p^2(\rho), \quad \hat{\sigma}_p^2(\rho) = \frac{1}{mk} \sum_{i=1}^m T_i(\bar{y}_i, \rho),$$

while the marginal log likelihood (12.13) when profiled gives

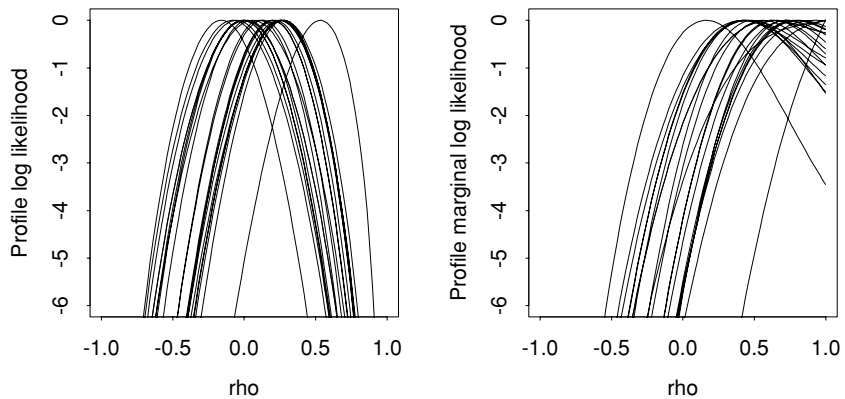
$$\max_{\sigma^2} \ell_m(\rho, \sigma^2) = -\frac{m}{2} \log \left\{ \frac{k - (k - 2)\rho}{1 + \rho} \right\} - \frac{m(k - 1)}{2} \log \hat{\sigma}_m^2(\rho),$$

where $\hat{\sigma}_m^2(\rho) = k\hat{\sigma}_p^2(\rho)/(k - 1)$. As $\hat{\sigma}_m^2(\rho) > \hat{\sigma}_p^2(\rho)$, there is an upward bias correction like that in Example 12.9. Let $\hat{\rho}$ and $\hat{\rho}_m$ denote the estimators from maximizing the usual and marginal likelihoods for ρ .

Figure 12.3 shows the profile log likelihood and the profile marginal log likelihood for 20 samples with $m = 10$, $k = 5$ and $\rho = 0.7$. Most of the ordinary maximum likelihood estimates $\hat{\rho}$ are much smaller than ρ , but the marginal maximum likelihood estimator $\hat{\rho}_m$ is much less downwardly biased, as is confirmed by further simulation. Some values of $\hat{\rho}_m$ equal the upper limit of $\rho = 1$, so the small-sample distributions of $\hat{\rho}_m$ and the marginal likelihood ratio statistic will be poorly approximated by asymptotic results. ■

Despite its artificiality, this example illustrates how estimation using restricted likelihood can give substantial bias corrections, and it is generally preferable to ordinary likelihood for variance estimation in complex models.

Figure 12.3 Inference for correlation parameter in short time series with $\rho = 0.7$. Left: profile log likelihoods for 20 samples consisting of $m = 10$ series of length $k = 5$. Right: profile marginal log likelihoods for ρ , for the same datasets. Maximum marginal likelihood estimators are much less biased than ordinary maximum likelihood estimators.



Regression-scale model

Consider the linear regression model $Y = X\beta + e^\tau \varepsilon$, where the errors $\varepsilon_1, \dots, \varepsilon_n$ are independent, the $n \times p$ design matrix X has rank p and rows $x_1^\tau, \dots, x_n^\tau$, τ is scalar and β is a $p \times 1$ vector of parameters. Exact results for this linear regression model were discussed in Chapters 8 and 9 for normal errors, but are typically unavailable otherwise. Accurate approximations to distributions needed for inference may be found, however. Let d be the negative log density of ε , and note that the log likelihood is

$$\ell(\beta, \tau; y) = -n\tau - \sum_{j=1}^n d\{e^{-\tau}(y_j - x_j^\tau \beta)\}.$$

The minimal sufficient statistic is typically the full set of responses $Y_1, \dots, Y_n$. Differentiation of the log likelihood shows that the maximum likelihood estimators $\hat{\beta}$ and $\hat{\tau}$ are determined by the score equations

$$\sum_{j=1}^n x_j d'(A_j) = 0, \quad n - \sum_{j=1}^n A_j d'(A_j) = 0, \quad (12.14)$$

where $d'(u)$ is the derivative of $d(u)$ with respect to u , and $A_j = e^{-\hat{\tau}}(Y_j - x_j^\tau \hat{\beta})$, for $j = 1, \dots, n$. Evidently $A = (A_1, \dots, A_n)^\tau$ is distribution constant, and as $\hat{\beta}$ and $\hat{\tau}$ are functions of the minimal sufficient statistic, A is in general also ancillary. Its distribution is degenerate, with $p + 1$ constraints imposed by (12.14), so just $n - p - 1$ of the A_j are needed to determine them all. Let A_0 denote such a subset, say, $A_0 = (A_1, \dots, A_{n-p-1})$.

We now consider how conditional tests and confidence intervals may be derived for an element of β , say β_1 without loss of generality. The idea is to construct a marginal density that depends only on β_1 , by using the joint density of $\hat{\beta}$ and $\hat{\tau}$ conditional on A to obtain

$$f(u_1, u_2 | a; \beta, \tau) = f(\hat{\beta}, \hat{\tau} | a; \beta, \tau) \left| \frac{\partial(\hat{\beta}, \hat{\tau})}{\partial(u_1, u_2)} \right|,$$

where $U_1 = e^{-\hat{\tau}}(\hat{\beta}_1 - \beta_1)$, U_2 denotes the $p \times 1$ vector with elements $\hat{\tau} - \tau$ and $e^{-\hat{\tau}}(\hat{\beta}_{-1} - \beta_{-1})$, and β_{-1} denotes $(\beta_2, \dots, \beta_p)^\tau$. We shall see that U_1 and U_2 are

pivots, and that the distribution of U_1 conditional on the ancillary statistic A forms the basis of conditional tests and confidence intervals for β_1 . These are readily obtained using p^* -type approximations.

As a preliminary step, we write the log likelihood as

$$\ell(\beta, \tau; y) = \ell \{e^{-\hat{\tau}}(\beta - \hat{\beta}), \tau - \hat{\tau}; a\} - n\hat{\tau}; \quad (12.15)$$

its maximized value is $\ell(\hat{\beta}, \hat{\tau}; a) = -n\hat{\tau} - \sum_{j=1}^n d(a_j)$. For later use define

$$\begin{aligned} h_a(u_1, u_2) &= \ell(0, 0; a) - \ell(u_1, u_2; a) \\ &= \ell(0, 0; a) - \ell \{e^{-\hat{\tau}}(\beta - \hat{\beta}), \tau - \hat{\tau}; a\}, \end{aligned}$$

and observe that its second derivative matrix with respect to (u_1, u_2) has determinant of form $e^{-2\hat{\tau}} H_a(u_1, u_2)$, and

$$\frac{\partial h_a(u_1, u_2)}{\partial u_1} = \frac{\partial \ell(\beta, \tau; y)}{\partial \beta_1}.$$

We now compute the density of $(\hat{\beta}, \hat{\tau})$ conditional on A . An extension of the argument giving (5.19) shows that the Jacobian for the transformation $y \mapsto (\hat{\beta}, \hat{\tau}, a_0)$ may be written $e^{(n-p)\hat{\tau}} J(a)$, so

$$\begin{aligned} f(\hat{\beta}, \hat{\tau}, a_0; \beta, \tau) &= J(a) \exp \{(n-p)\hat{\tau} + \ell(\beta, \tau; y)\} \\ &= J(a) \exp [-p\hat{\tau} + \ell \{e^{-\hat{\tau}}(\beta - \hat{\beta}), \tau - \hat{\tau}; a\}]. \end{aligned}$$

Hence

$$f(\hat{\beta}, \hat{\tau} | a; \beta, \tau) = \frac{\exp [-p\hat{\tau} + \ell \{e^{-\hat{\tau}}(\beta - \hat{\beta}), \tau - \hat{\tau}; a\}]}{\int \exp [-p\hat{\tau} + \ell \{e^{-\hat{\tau}}(\beta - \hat{\beta}), \tau - \hat{\tau}; a\}] d\hat{\beta} d\hat{\tau}},$$

the domain of integration being $\mathbb{R}^{p+1}$. We now change variables in the integral from $(\hat{\beta}, \hat{\tau})$ to (u_1, u_2) , with Jacobian $\exp(p\hat{\tau})$. Thus the integral may be written

$$\int \exp \{\ell(0, 0; a) - h_a(u_1, u_2)\} du_1 du_2$$

and this equals

$$(2\pi)^{(p+1)/2} \left| \frac{\partial^2 h_a(0, 0)}{\partial u \partial u^T} \right|^{-1/2} e^{\ell(0, 0; a)} \{1 + O(n^{-1})\}$$

by Laplace approximation, the $O(n^{-1})$ term depending only on a , and the conditional density $f(\hat{\beta}, \hat{\tau} | a; \beta, \tau)$ equals

$$\bar{c}(a)(2\pi)^{-(p+1)/2} \left| \frac{\partial^2 h_a(0, 0)}{\partial u \partial u^T} \right|^{1/2} \hat{\tau}^{-p} \exp [\ell \{e^{-\hat{\tau}}(\beta - \hat{\beta}), \tau - \hat{\tau}; a\} - \ell(0, 0; a)],$$

where $\bar{c}(a) = 1 + O(n^{-1})$ depends only on a . On changing variables to u_1 and u_2 , we have

$$f(u_1, u_2 | a; \beta, \tau) = \bar{c}(a)(2\pi)^{-(p+1)/2} \left| \frac{\partial^2 h_a(0, 0)}{\partial u \partial u^T} \right|^{1/2} \exp \{-h_a(u_1, u_2)\},$$

and as the the right-hand side does not depend on the parameters, $U_1(\beta_1)$ and $U_2(\beta_{-1}, \tau)$ are pivots. Note that the inferences below are not parametrization invariant, because the accuracy of Laplace approximation depends on the closeness of h to quadratic; taking $\hat{\tau} - \tau$ should give better results than the more obvious pivot $\hat{\sigma}/\sigma$.

Consider testing the hypothesis that β_1 takes value β_1^0 . If so, the observed value of U_1 is $u_1^0 = e^{-\hat{\tau}}(\hat{\beta}_1 - \beta_1^0)$, with corresponding tail probability

$$\Pr(U_1 \leq u_1^0 \mid A = a) = \bar{c}(a) \frac{|\partial^2 h_a(0, 0)/\partial u \partial u^\top|^{1/2}}{(2\pi)^{(p+1)/2}} \int_{-\infty}^{u_1^0} du_1 \int du_2 e^{-h_a(u_1, u_2)},$$

the inner integral being over $\mathbb{R}^p$. This expression has form (11.32), so

$$\Pr(U_1 \leq u_1^0 \mid A = a) = \Phi\{r^*(\beta_1^0)\} \{1 + O(n^{-1})\}, \quad (12.16)$$

where $r^*(\beta_1^0) = r(\beta_1^0) + r(\beta_1^0)^{-1} \log\{v(\beta_1^0)/r(\beta_1^0)\}$, and

$$r(\beta_1^0) = \text{sign}(u_1^0) \{2h_a(u_1^0, \tilde{u}_{20})\}^{1/2},$$

$$v(\beta_1^0) = \frac{\partial h_a(u_1^0, \tilde{u}_{20})}{\partial u_1} \left| \frac{\partial^2 h_a(u_1^0, \tilde{u}_{20})}{\partial u_2 \partial u_2^\top} \right|^{1/2} \left| \frac{\partial^2 h_a(0, 0)}{\partial u \partial u^\top} \right|^{-1/2},$$

where $\tilde{u}_{20}$ is the value of u_2 that maximizes $h_a(u_1^0, u_2)$ with $u_1^0 = e^{-\hat{\tau}}(\hat{\beta}_1 - \beta_1^0)$ fixed. It is straightforward to check that in terms of the log likelihood and its derivatives,

$$r(\beta_1^0) = \text{sign}(\hat{\beta}_1 - \beta_1^0) [2\{\ell(\hat{\beta}, \hat{\tau}; y) - \ell(\hat{\beta}^0, \hat{\tau}^0; y)\}]^{1/2},$$

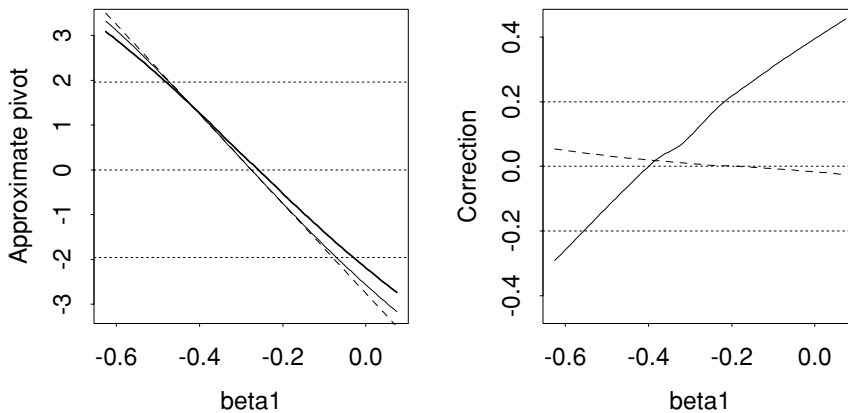
$$v(\beta_1^0) = \frac{\partial \ell(\hat{\beta}^0, \hat{\tau}^0; y)}{\partial \beta_1} \frac{|J_{\lambda\lambda}(\hat{\beta}^0, \hat{\tau}^0; y)|^{1/2}}{|J(\hat{\beta}^0, \hat{\tau}^0; y)|^{1/2}} \times \frac{|J(\hat{\beta}^0, \hat{\tau}^0; y)|^{1/2}}{|J(\hat{\beta}, \hat{\tau}; y)|^{1/2}}, \quad (12.17)$$

say, where $\hat{\beta}^0 = (\beta_1^0, \hat{\beta}_{-1}^0)$, $\hat{\beta}_{-1}^0$ and $\hat{\tau}^0$ are the maximum likelihood estimates of β_{-1} and τ when $\beta_1 = \beta_1^0$, $J(\beta, \tau; y)$ is the observed information matrix for β and τ , and $J_{\lambda\lambda}(\beta, \tau; y)$ is the observed information matrix corresponding to $\lambda = (\beta_{-1}, \tau)$ only. Note that $r(\beta_1^0) = v(\beta_1^0) = 0$ when $\beta_1^0 = \hat{\beta}_1$; it turns out that $r^*(\beta_1^0)$ has a finite limit as $\beta_1^0 \rightarrow 0$.

Our argument has established that the conditional tail probability (12.16) depends on the signed likelihood ratio statistic $r(\beta_1^0)$ and a modified score statistic $v(\beta_1^0)$ for testing the hypothesis $\beta_1 = \beta_1^0$. Expression (12.17) may be written as a product, γC , say, where the first term is a score statistic standardized by its standard error and the second contains information matrix determinants. Then $r^*(\beta_1^0)$ may be decomposed as

$$r^* = r + r^{-1} \log(\gamma/r) + r^{-1} \log C = r + r_{\text{INF}} + r_{\text{NP}}, \quad (12.18)$$

say, where dependence on β_1^0 has been suppressed. The discussion around (12.9) suggests that r_{INF} should be regarded as a correction that improves the normal approximation to the distribution of the signed likelihood ratio statistic. The presence of λ introduces the term r_{NP} , large values of which indicate that strong correction for nuisance parameters is required. Although r_{INF} is typically small in applications, the nuisance parameter correction r_{NP} can be substantial.

**Figure 12.4**

Small-sample inference on parameter β_1 for partial turnkey guarantee for nuclear plant data, using regression-scale model with t_5 errors. Left: approximate pivots $r(\beta_1)$ (solid), $r^*(\beta_1)$ (heavy), and $(\hat{\beta}_1 - \beta_1)/v_{11}^{1/2}$ (dashes) as functions of β_1 . Horizontal dotted lines are at 0 and ± 1.96 , so their intersections with the diagonal lines show the limits of approximate 0.95 confidence intervals. Right: small-sample corrections r_{NP} (solid) and r_{INF} (dashes) as functions of β_1 . The horizontal lines at ± 0.2 give a crude rule of thumb for 'large' corrections.

The limits of an equi-tailed $(1 - 2\alpha)$ conditional confidence interval for β_1 are the values β_1^+ , β_1^- that satisfy

$$\Pr \{ U_1 \leq e^{-\hat{\tau}}(\hat{\beta}_1 - \beta_1) | a \} \doteq \Phi \{ r^*(\beta_1) \} = \alpha, 1 - \alpha.$$

The likelihood ratio confidence limits are obtained by replacing $r^*(\beta_1)$ with $r(\beta_1)$ in this expression, so computation of $v(\beta_1)$ for use in $r^*(\beta_1)$ is the only addition needed for small-sample conditional inference. Routines for large-sample likelihood inference on the model using $r(\beta_1)$ compute most of the quantities involved for small-sample conditional inferences using $r^*(\beta_1)$, so rather little extra work is needed.

Example 12.12 (Nuclear plant data) To illustrate the effect of these small-sample modifications, we fit the model with six covariates chosen in Example 8.31 to the data in Table 8.13. We focus on the effect of the partial turnkey guarantee, PT, the coefficient of which under the normal model is -0.266 with standard error 0.144 , giving an exact 0.95 confidence interval $(-0.46, 0.01)$. The evidence for inclusion of this effect is somewhat marginal under the normal model, and it is interesting to see the effect of using a t_5 error distribution instead. Computations that generalize those in Example 12.7 lead to the results shown in Figure 12.4, the left panel of which shows how $r(\beta_1)$ and $r^*(\beta_1)$ depend on the parameter β_1 for the partial turnkey effect. The 0.95 confidence intervals based on these pivots are respectively $(-0.47, -0.07)$ and $(-0.48, -0.03)$, while that based on normal approximation to the maximum likelihood estimator $\hat{\beta}_1$ is $(-0.47, -0.08)$. The more accurate interval based on the small-sample approximation is rather wider than the others, and the right panel of the figure shows that this is primarily due to the nuisance parameter adjustment.

None of the intervals differs much from that found with normal errors, however, giving some reassurance that conclusions found using the normal model are valid more broadly, at least for this example. ■

Exercises 12.2

- 1 Suppose that $Y_1, \dots, Y_n$ are independent Poisson variables with means $\psi\pi_j$, where $0 < \pi_j < 1$ and $\sum \pi_j = 1$. Find a marginal likelihood for ψ based on $Y_1, \dots, Y_n$, and show that no information about ψ is lost by using the marginal likelihood rather than the full likelihood.
- 2 Verify the argument in Example 12.10 by writing down the partial likelihood for observations $0 < Y_4 < Y_1^+ < Y_3^+ < Y_5 < Y_2^+$ from the proportional hazards model, and then deriving it by the reasoning in the example.
- 3 Use the spectral decomposition to show that an $n \times (n - p)$ matrix B exists such that $B^T B = I_n - H$, and $BB^T = I_{n-p}$, where $H = X(X^T X)^{-1} X^T$ is the projection matrix for a linear model with $n \times p$ design matrix X of rank p .
- 4 Show that the maximum likelihood estimator of σ^2 based on (12.13) is given by

$$\hat{\sigma}_m^2 = \frac{1}{n - p} (y - X\hat{\beta}_\psi)^T \Upsilon (y - X\hat{\beta}_\psi), \quad \hat{\beta}_\psi = (X^T \Upsilon X)^{-1} X^T \Upsilon y.$$

Deduce that when Zb does not appear in (12.10), the restricted maximum likelihood estimator of σ^2 is $(y - X\hat{\beta})^T (y - X\hat{\beta}) / (n - p)$.

- 5 Find marginal likelihoods for σ^2 and for $\psi = (\beta_0, \sigma^2)$ in Example 8.10.
- 6 Check the details of Example 12.11.
- 7 Verify the p^* formula (12.7) for the regression-scale model, replacing $(2\pi)^{-1/2}$ by $(2\pi)^{-(p+1)/2}$.
- 8 (a) Express the quantities $r(\beta_1^0)$ and $v(\beta_1^0)$ defined after (12.16) in terms of the log likelihood and its derivatives.
 (b) Compute the modified signed likelihood ratio statistic for inference on τ .
 (c) Find $r^*(\beta_1)$ for the regression-scale model when $d(u) = u^2/2$.

12.3 Conditional Inference

12.3.1 Exact conditioning

Thus far this chapter has focused on the use of conditioning to ensure the relevance of tests and confidence intervals. In exponential family models there is typically no ancillary statistic and conditioning has the somewhat different purpose of removing a nuisance parameter from consideration; see Sections 10.4.2, 10.5, and 10.8.2. If the model density may be factorized as

$$f(y; \psi, \lambda) \propto f(t_1 \mid t_2; \psi) f(t_2; \psi, \lambda), \quad (12.19)$$

then although both terms on the right contain information on ψ , it may not be worthwhile to try and extract it from the second term. Moreover in Section 7.3.3 we saw that similar critical regions for tests on ψ must be based on the conditional density of T_1 given T_2 . These considerations suggest that we restrict consideration to this density, which we treat as a conditional likelihood for ψ . Exact inference is typically infeasible, however.

Example 12.13 (Logistic regression) Let $Y_1, \dots, Y_n$ be independent binary variables having $p \times 1$ covariate vectors x_j and satisfying a logistic regression model.

The minimal sufficient statistic $\sum x_j Y_j = S = (S_1, \dots, S_p)^T$ has joint density

$$\Pr(S_1 = s_1, \dots, S_p = s_p; \beta) = \frac{c(s_1, \dots, s_p) \exp(s_1 \beta_1 + \dots + s_p \beta_p)}{\sum_{j=1}^n \left\{ 1 + \exp(x_j^T \beta) \right\}},$$

found by summing the joint density (10.23) of the Y_j over all $c(s_1, \dots, s_p)$ binary sequences of length n that yield the same value of S as did the data. If β_p is taken as the parameter of interest and the other β_j are treated as nuisance parameters, then they do not appear in the conditional density

$$\Pr(S_p = s_p \mid S_1 = s_1, \dots, S_{p-1} = s_{p-1}; \beta_p) = \frac{c(s_1, \dots, s_p) e^{s_p \beta_p}}{\sum c(s_1, \dots, s_{p-1}, u) e^{u \beta_p}},$$

where the sum is over the possible values of u for which $s_1, \dots, s_{p-1}$ are fixed. Exact tests and confidence intervals may be obtained by adapting the argument on page 495, but rely on ready computation of the combinatorial coefficients. In principle these may be obtained by considering the coefficients of $w_1^{s_1} \cdots w_p^{s_p}$ in the expansion of the generating function

$$\prod_{j=1}^n (1 + w_1^{x_{j1}} \cdots w_p^{x_{jp}}),$$

where x_{jr} is the r th element of x_j . Recourse to computer algebra or special algorithms is needed for all but the simplest problems, however, and typically the computational burden puts exact inference out of reach. ■

In subsequent sections we discuss accurate analytical approximations to exact conditional inferences, but we first illustrate how Markov chain Monte Carlo simulation (Section 11.3.3) may be used to explore a conditional sample space.

Example 12.14 (Several 2×2 tables) Consider n independent 2×2 tables containing the numbers of successes R_{1j} and R_{0j} in m_{1j} and m_{0j} independent Bernoulli trials with success probabilities

$$\pi_{1j} = \frac{e^{\lambda_j + \psi_j}}{1 + e^{\lambda_j + \psi_j}} \quad \text{and} \quad \pi_{0j} = \frac{e^{\lambda_j}}{1 + e^{\lambda_j}}, \quad j = 1, \dots, n.$$

The overall joint density is

$$\prod_{j=1}^n \binom{m_{1j}}{r_{1j}} \binom{m_{0j}}{r_{0j}} \frac{e^{r_{1j}(\lambda_j + \psi_j)}}{(1 + e^{\lambda_j + \psi_j})^{m_{1j}}} \frac{e^{r_{0j} \lambda_j}}{(1 + e^{\lambda_j})^{m_{0j}}}.$$

Suppose we wish to test whether the log odds ratios are equal for each table, that is $\psi_1 = \dots = \psi_n = \psi$, where ψ is unknown. Let $W = w(R)$ be a statistic constructed to test this against the alternative of unequal ψ_j . Under the null hypothesis the statistics $S_j = R_{1j} + R_{0j}$ and $T = R_{11} + \dots + R_{1n}$ are associated with λ_j and ψ , and as we wish to test homogeneity of the log odds ratios regardless of the values of $\lambda_1, \dots, \lambda_n, \psi$, we should use the conditional distribution of W given the values of

$S_1, \dots, S_n, T$ that were actually observed, namely $s_1, \dots, s_n, t$. It is straightforward to establish that

$$\Pr(R \mid S, T) = \frac{\prod_{j=1}^n \binom{m_{1j}}{r_{1j}} \binom{m_{0j}}{s_j - r_{1j}}}{\sum \prod_{j=1}^n \binom{m_{1j}}{u_j} \binom{m_{2j}}{s_j - u_j}} = C^{-1} \prod_{j=1}^n c_j(r_{1j}), \quad (12.20)$$

say, where the sum is over the set

$$\mathcal{U} = \{(u_1, \dots, u_n) : u_{j,-} \leq u_j \leq u_{j,+}, u_j \in \mathbb{Z}, u_1 + \dots + u_n = t\},$$

with $u_{j,-} = \max(0, s_j - m_{0j})$ and $u_{j,+} = \min(m_{1j}, s_j)$. If the 2×2 tables were stacked on top of one another, then this conditioning would amount to fixing all three margins of the stack. We would like to use (12.20) to calculate the tail probability $\Pr\{w(R) \geq w \mid S = s, T = t\}$, but enumeration of $\mathcal{U}$ in order to find the normalising constant in (12.20) is typically difficult. A simplifying feature is that any 2×2 table for which s_j equals zero or $m_{0j} + m_{1j}$ is conditionally constant and can be ignored.

One possibility is to use the Metropolis–Hastings algorithm. The idea is to update the stack of 2×2 tables with contents r to have contents r' , while holding its margins constant. Although it is hard to simulate directly from (12.20), it is straightforward to generate from the conditional distribution of R_{1i} given $R_{1j} + R_{1i}$, which is univariate with density

$$\Pr(R_{1i} = r \mid R_{1j} + R_{1i} = v) = \frac{c_i(r)c_j(v-r)}{\sum c_i(u)c_j(v-u)},$$

where the sum is over values of $R_{1i} = u$ for which $R_{1j} + R_{1i} = v$ is possible. The algorithm starts with original data r and repeats the following steps for $k = 1, \dots, K$.

1. For $l = 1, \dots, L$,
 - select two tables in the current stack at random, say i and j ;
 - generate proposal data r' by leaving the remaining 2×2 tables intact and generating a value r'_{1i} from the conditional distribution of R_{1i} given $R_{1i} + R_{1j}$; then set $r'_{1j} = r_{1i} + r_{1j} - r'_{1i}$.
 - Set $r \leftarrow r'$ with probability

$$p = \min \left\{ 1, \frac{\pi(r')q(r \mid r')}{\pi(r)q(r' \mid r)} \right\}. \quad (12.21)$$

2. Calculate $w_k^* = w(r)$ based on the current data r .

The estimated P-value is then $\{\sum I(w_k^* \geq w_{\text{obs}}) + 1\}/(K + 1)$, where w_{obs} is the observed value of $w(R)$.

The target density is

$$\pi(r) \propto \prod_j c_j(r_{1j})$$

and the transition density for the Markov chain is

$$q(r' \mid r) \propto c_i(r'_{1i})c_j(r_{1i} + r_{1j} - r'_{1i}) \sum_u c_i(u)c_j(r_{1i} + r_{1j} - u),$$

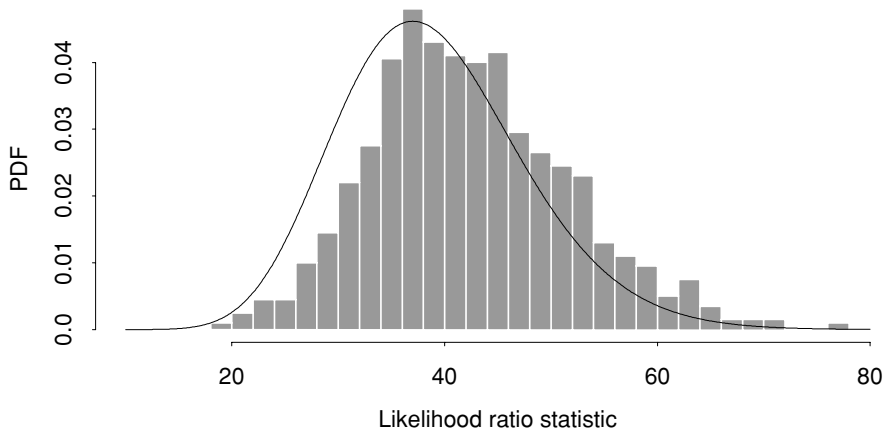


Figure 12.5 Simulation approximation to conditional distribution of test statistic W for homogeneity of parameters ψ_j for the ulcer data (Table 10.11). The line is the asymptotic χ^2_{39} density.

so the ratio in (12.21) is

$$\frac{\pi(r')q(r | r')}{\pi(r)q(r' | r)} = \frac{c_i(r'_{1i})c_j(r'_{1j})c_i(r_{1i})c_j(r_{1j})}{c_i(r_{1i})c_j(r_{1j})c_i(r'_{1i})c_j(r'_{1j})} = 1.$$

Thus the proposal r' is always accepted.

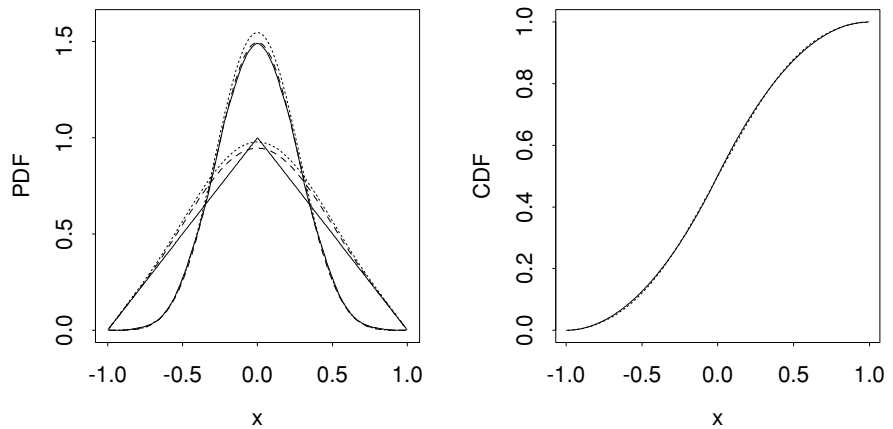
In order to assess whether the parameters ψ_j for recurrent bleeding vary across the 41 studies in Table 10.11, we apply this algorithm to the 40 non-degenerate tables; the last one is conditionally constant. We take W to be the likelihood ratio statistic for testing the null hypothesis. Its observed value is $w = 177.6$ on 39 degrees of freedom, which is highly significant relative to its asymptotic χ^2_{39} distribution. Values w^* of W^* calculated every $L = 100$ steps using the algorithm above were uncorrelated, and are compared with the asymptotic distribution in Figure 12.5. The evidence that the ψ_j vary across the tables is overwhelming relative to either distribution, but as the distributions are rather different, conflict between asymptotic and exact P-values could arise for other such sets of data; then the P-value based on W^* would be preferred. Such a P-value is called *Monte Carlo exact*: it would be exact if the number of Monte Carlo replicates was infinite. ■

Algorithms of this type provide powerful tools for testing hypotheses in contingency tables with low counts, and can be adapted to provide Monte Carlo approximations to conditional likelihoods. A difficulty with exact conditioning, however, is that in some cases the inference depends very sharply on the conditioning event, giving likelihoods, significance levels and so forth that are highly sensitive to the exact data values. This raises the question whether some form of approximate conditioning would be more stable. Below we outline how conditional inferences in exponential families may be approximated using saddlepoint methods.

12.3.2 Saddlepoint approximation

Saddlepoint approximation to density and distribution functions is the basis of many highly accurate small-sample procedures. In this section we describe informally the

Figure 12.6 Saddlepoint approximations to the density and distribution functions of an average of n $U(-1, 1)$ variables. Left panel: exact density functions (solid) and their basic (dots) and normalized (dashes) approximations for $n = 2, 5$; the peakier curves are for $n = 5$. Right panel: exact distribution function (solid) and its approximation (dots) for $n = 2$.



underlying ideas. Initially we state the basic approximations, leaving their justification to the end of the section.

Let $\bar{X}$ denote the average of a random sample of continuous scalar random variables $X_1, \dots, X_n$, each having cumulant-generating function $K(u)$. Then the *saddlepoint approximation* to the density of $\bar{X}$ at x is

$$f_{\bar{X}}(x) \doteq \left\{ \frac{n}{2\pi K''(\tilde{u})} \right\}^{1/2} \exp [n \{K(\tilde{u}) - \tilde{u}x\}], \quad (12.22)$$

where $\tilde{u} = \tilde{u}(x)$, known as the saddlepoint, is the unique value of u satisfying the *saddlepoint equation* $K'(u) = x$, and $K'(u)$ and $K''(u)$ are the first and second derivatives of $K(u)$ with respect to u . There is a corresponding approximation to the cumulative distribution function of $\bar{X}$, namely

$$\Pr\{\bar{X} \leq x\} \doteq \Phi\{r^*(x)\}, \quad (12.23)$$

where $r^*(x) = r(x) + r(x)^{-1} \log\{v(x)/r(x)\}$ depends on

$$r(x) = \text{sign}(\tilde{u}) [2n \{\tilde{u}x - K(\tilde{u})\}]^{1/2}, \quad v(x) = \tilde{u} \{nK''(\tilde{u})\}^{1/2}.$$

Calculation of (12.22) and (12.23) requires knowledge of $K(u)$ and computation of $\tilde{u}$ for each x of interest, but such approximations are often extremely accurate far into the tails of the density of $\bar{X}$. A more accurate density approximation can be obtained by renormalizing (12.22), that is, dividing by its integral, which is usually obtained numerically.

Example 12.15 (Uniform distribution) If X has the uniform distribution on $(-1, 1)$, then $K(u) = \log\{\sinh(u)/u\}$. The left panel of Figure 12.6 compares the exact density function of $\bar{X}$ with its saddlepoint approximations when $n = 2$ and $n = 5$. Numerical integrals of the density approximations are 1.097 and 1.034, and the renormalized densities are also shown; that for $n = 5$ seems essentially exact. The distribution function approximation (12.23) is very accurate even in the extreme situation $n = 2$, although (12.22) fails to capture the cusp in the density at the origin. ■

These basic approximations can be generalized in various ways. For our purpose the most useful is to the situation where the X_j are vectors of length p , in which case u is a $p \times 1$ vector. Then (12.22) extends to

$$f_{\bar{X}}(x) \doteq \left\{ \frac{n^p}{(2\pi)^p |K''(\tilde{u})|} \right\}^{1/2} \exp \left[n \{ K(\tilde{u}) - \tilde{u}^T x \} \right], \quad (12.24)$$

where now the $p \times 1$ saddlepoint $\tilde{u}$ solves the $p \times 1$ system of equations

$$K'(\tilde{u}) = \frac{\partial K(\tilde{u})}{\partial u} = x,$$

and $K''(u) = \partial^2 K(u) / \partial u \partial u^T$ is the $p \times p$ matrix of second derivatives of $K(u)$.

Let $X^T = (X_1, X_2^T)$, where X_2 has dimension $(p-1) \times 1$, and split u^T into (u_1, u_2^T) conformably with X^T ; both X_1 and u_1 are scalar. Then the marginal density of $\bar{X}_2$ at x_2 is obtained by saddlepoint approximation using the marginal cumulant-generating function $K(0, u_2)$ of X_2 , and is

$$f_{\bar{X}_2}(x_2) \doteq \left\{ \frac{n^{p-1}}{(2\pi)^{p-1} |K''_{22}(\tilde{u}_0)|} \right\}^{1/2} \exp \left[n \{ K(\tilde{u}_0) - (0, x_2^T) \tilde{u}_0 \} \right], \quad (12.25)$$

where $\tilde{u}_0^T = (0, \tilde{u}_2^T)$ is the solution to the $(p-1) \times 1$ system of equations $\partial K(0, u_2) / \partial u_2 = x_2$, and K''_{22} is the $(p-1) \times (p-1)$ corner of K'' corresponding to u_2 . Division of (12.24) by (12.25) gives an approximation to the conditional density $f_{\bar{X}_1 | \bar{X}_2}(x_1 | x_2)$, the *double saddlepoint approximation*

$$\left(\frac{n}{2\pi} \right)^{1/2} \frac{|K''_{22}(\tilde{u}_0)|^{1/2}}{|K''(\tilde{u})|^{1/2}} \exp \left[n \{ K(\tilde{u}) - \tilde{u}^T x - K(\tilde{u}_0) + (0, x_2^T) \tilde{u}_0 \} \right], \quad (12.26)$$

corresponding to which is the distribution function approximation

$$\Pr(\bar{X}_1 \leq x_1 | \bar{X}_2 = x_2) \doteq \Phi \{ r^*(x_1) \}, \quad (12.27)$$

where again $r^*(x_1)$ equals $r(x_1) + r(x_1)^{-1} \log \{ v(x_1) / r(x_1) \}$, but now with

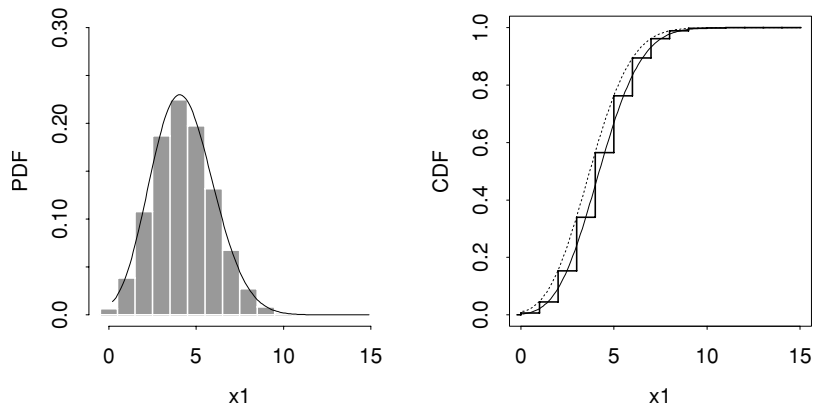
$$\begin{aligned} r(x_1) &= \text{sign}(\tilde{u}_1) \left[2n \{ K(\tilde{u}_0) - (0, x_2^T) \tilde{u}_0 \} - n \{ K(\tilde{u}) - \tilde{u}^T x \} \right]^{1/2}, \\ v(x_1) &= \tilde{u}_1 n^{1/2} |K''(\tilde{u})|^{1/2} / |K''_{22}(\tilde{u}_0)|^{1/2}. \end{aligned}$$

These formulae break down when $x_1 = E(X_1)$ and a different more complicated approximation is then available. In practice it is simplest to evaluate $r^*(x_1)$ for at a grid of values of x_1 excluding any too close to $E(X_1)$, and then to interpolate them using a spline or other numerical method.

Example 12.16 (Poisson distribution) Let V_1 and V_2 be independent Poisson variables with means λ_1 and λ_2 , and set $X_1 = V_1$, $X_2 = V_1 + V_2$. Then the joint cumulant-generating function of X_1 and X_2 is $K(u) = \lambda_1 \exp(u_1 + u_2) + \lambda_2 \exp(u_2)$, and the exact density of X_1 given $X_2 = x_2$ is binomial with probability $\pi = \lambda_1 / (\lambda_1 + \lambda_2)$ and denominator x_2 .

Figure 12.7 shows the conditional density and distribution of X_1 , when $\lambda_1 = 1$, $\lambda_2 = 2.5$, and $x_2 = 15$. The saddlepoint density gives a continuous approximation to the discrete binomial density of X_1 given X_2 , which it closely matches on the support

Figure 12.7 Double saddlepoint approximations to the density and distribution functions of $X_1 = V_1$ given $X_2 = V_1 + V_2 = 15$, where V_1 and V_2 are Poisson with means 1 and 2.5. Left panel: exact density function and its approximation. Right panel: exact distribution function (heavy) and its approximation without (solid) and with (dots) a continuity correction.



of X_1 . For $x_1 = 0, \dots, 15$, the distribution function approximation $\Phi\{r^*(x_1)\}$ agrees closely with

$$\Pr(X_1 \leq x_1 - 1 \mid X_2) + \frac{1}{2}\Pr(X_1 = x_1 \mid X_2),$$

a quantity akin to the mid- p significance level (10.28). Continuity correction applied by taking $\Phi\{r^*(x_1 + 1/2)\}$ approximates $\Pr(X_1 \leq x_1 \mid X_2)$ well. ■

In applications saddlepoint formulae are sometimes used simply by taking the cumulant-generating function for a variable of interest, and formally setting $n = 1$, as we shall do below.

The rest of this section sketches derivations of the approximations above and can be skipped on a first reading. No attempt is made at a rigorous treatment; the intention is to give some justification for the approximations, and the flavour of the manipulations involved.

Edgeworth series

Although they have practical disadvantages compared to saddlepoint approximations, Edgeworth series play a central role in theoretical discussions of small-sample inference. As we shall see below, they allow a relatively simple derivation of saddlepoint approximations.

Let $X_1, \dots, X_n$ be a random sample of continuous variables with cumulant-generating function $K(u)$ and finite cumulants κ_r , let $\rho_r = \kappa_r / \kappa_2^{r/2}$ denote the r th standardized cumulant, and let $Z_n = (S_n - n\kappa_1) / (n\kappa_2)^{1/2}$ denote the standardized version of $S_n = X_1 + \dots + X_n$. The large-sample distribution of Z_n may be found by noting that its cumulant-generating function is

$$u^2/2 + n^{-1/2}\rho_3 u^3/3! + n^{-1}\rho_4 u^4/4! + \dots$$

Thus its moment-generating function is

$$\exp(u^2/2) \left\{ 1 + \frac{1}{6}\rho_3 n^{-1/2} u^3 + \left(\frac{\rho_3^2}{72} + \frac{\rho_4}{24} \right) n^{-1} u^4 + O(n^{-3/2}) \right\}. \quad (12.28)$$

Francis Ysidro Edgeworth (1845–1926) was born in Longford, Ireland, studied in Dublin and Oxford, where he took a first-class degree in classics, and then became a London barrister. In 1880 he began a university career as a mathematician, eventually holding professorships of political economy in London and Oxford. His pioneering articles systematically applied ideas from the theory of errors to social science and economics.

All that remains when $n \rightarrow \infty$ is the leading term, which is the moment-generating function of the standard normal density. The continuity theorem and uniqueness of moment-generating functions then yield the limiting standard normal distribution of Z_n .

To find better finite-sample approximations to the density and distribution functions of Z_n , note that integration by parts gives

$$\int_{-\infty}^{\infty} e^{uz} (-1)^r \frac{d^r \phi(z)}{dz^r} dz = u^r e^{u^2/2}, \quad r = 0, 1, \dots$$

Letting $(-1)^r d^r \phi(z)/dz^r = \phi(z) H_r(z)$ determine the r th-order Hermite polynomial $H_r(z)$ gives

$$\begin{aligned} H_1(z) &= z, \quad H_2(z) = z^2 - 1, \quad H_3(z) = z^3 - 3z, \quad H_4(z) = z^4 - 6z^2 + 3, \\ H_5(z) &= z^5 - 10z^3 + 15z, \quad H_6(z) = z^6 - 15z^4 + 45z^2 - 15. \end{aligned}$$

Term by term inversion of (12.28) shows that the corresponding density is

$$f_{Z_n}(z) = \phi(z) \left[1 + \frac{\rho_3}{6n^{1/2}} H_3(z) + \frac{1}{n} \left\{ \frac{\rho_4}{24} H_4(z) + \frac{\rho_3^2}{72} H_6(z) \right\} + O(n^{-3/2}) \right], \quad (12.29)$$

inversion of which gives the first three terms of the cumulant-generating function of Z_n . Integration of (12.29) gives the corresponding distribution function,

$$F_{Z_n}(z) = \Phi(z) - \phi(z) \left[\frac{\rho_3}{6n^{1/2}} H_2(z) + \frac{1}{n} \left\{ \frac{\rho_4}{24} H_3(z) + \frac{\rho_3^2}{72} H_5(z) \right\} + O(n^{-3/2}) \right]. \quad (12.30)$$

The leading terms of the *Edgeworth expansions* (12.29) and (12.30) give the standard normal approximation for Z_n , with subsequent terms giving improvements, based respectively on the skewness of Z_n and on a combination of its skewness and kurtosis. Higher-order terms depend on further cumulants of the X_j .

The expansions (12.29) and (12.30) are more useful for theoretical development than for applications, because there is no reason for (12.29) to remain positive or for (12.30) to be increasing in z , or indeed even for it to lie between zero and one, whereas the saddlepoint formula (12.22) is guaranteed to give a positive density approximation. On the other hand saddlepoint approximation requires the entire cumulant-generating function, while (12.29) and (12.30) use only first few cumulants and so are more readily calculated.

When the density for Z_n is evaluated at $z = 0$, the series in (12.29) contains only powers of n^{-1} , because the odd powers of $n^{-1/2}$ depend on odd Hermite polynomials, all of which vanish at $z = 0$.

Derivation of saddlepoint approximation

To derive (12.22) we first embed the density f_X of X in the exponential family

$$f_X(x; u) = \exp\{xu - K(u)\} f_X(x),$$

where K is the cumulant-generating function of X . Here u plays the role of a parameter, and the new density $f_X(x; u)$ is an exponential tilt of f_X . Under this new model

the density of S_n may be written

$$f_{S_n}(s; u) = \exp\{su - nK(u)\} f_{S_n}(s), \quad (12.31)$$

and its cumulant-generating function as $K_{S_n}(t) = n\{K(t+u) - K(u)\}$. Thus with parameter u the mean and variance of S_n under the new density are $nK'(u)$ and $nK''(u)$. Expression (12.31) gives

$$f_{S_n}(s) = \exp\{nK(u) - su\} f_{S_n}(s; u)$$

for all u . We now replace $f_{S_n}(s; u)$ in this expression by its Edgeworth expansion, but with the value of u chosen so that the tilted density $f_{S_n}(s; u)$ has mean s . Then $nK'(\tilde{u}) = s$, and the first term of the Edgeworth series is $\{2\pi nK''(\tilde{u})\}^{-1/2}$. The resulting approximation to the density of S_n is

$$f_{S_n}(s) = \{2\pi nK''(\tilde{u})\}^{-1/2} \exp\{nK(\tilde{u}) - s\tilde{u}\} \{1 + O(n^{-1})\},$$

where the order of the error follows from the fact that at its mean the Edgeworth series contains only powers of n^{-1} . A change of variables $s \mapsto x = s/n$ gives the leading term of (12.22), with subsequent terms found by retaining more terms of the Edgeworth series.

The argument leading to (12.23) starts by integrating (12.22), giving

$$\Pr(\overline{X} \leq x_0) \doteq \int_{-\infty}^{x_0} \left\{ \frac{n}{2\pi K''(\tilde{u})} \right\}^{1/2} \exp[n\{K(\tilde{u}) - \tilde{u}x\}] dx,$$

where $\tilde{u}$ is a function of the variable of integration through $K'(\tilde{u}) = x$. A change of variable from x to $\tilde{u}$, using $dx/d\tilde{u} = K''(\tilde{u})$, gives

$$\Pr(\overline{X} \leq x_0) \doteq \int_{-\infty}^{u_0} \left\{ \frac{nK''(\tilde{u})}{2\pi} \right\}^{1/2} \exp[n\{K(\tilde{u}) - \tilde{u}K'(\tilde{u})\}] d\tilde{u},$$

which is of form (11.30); here u_0 satisfies $K'(u_0) = x_0$. In this case the approximation (11.31) takes the form (12.23). Detailed accounting shows that under broad conditions the error in (12.23) is relative, of size $O(n^{-1})$ in large deviation regions, where $x_0 - E(X)$ is $O(1)$, and is only $O(n^{-3/2})$ for moderate deviation regions, in which $x_0 - E(X)$ is $O(n^{-1/2})$. The large-deviation property leads to the extraordinary accuracy of the approximations, which have low relative error far into the tails of the distribution of $\overline{X}$.

Discrete distributions are also important in practice. Suppose that the X_j take values in the lattice $a + bk$, where a and b are constants and $k = 0, 1, \dots$. Then $\overline{X}$ takes values in the lattice $a + bk/n$. The saddlepoint approximation for the density of $\overline{X}$ at these values is unchanged, and its error remains relative and of $O(n^{-1})$, but in the cumulative distribution function approximation $v(x) = \tilde{u} \{nK''(\tilde{u})\}^{1/2}$ is replaced by $b^{-1} \{1 - \exp(-b\tilde{u})\} \{nK''(\tilde{u})\}^{1/2}$; note that this produces the continuous version when $b \rightarrow 0$. The error in this approximation is $O(n^{-1})$. A continuity-corrected approximation to the quantity $\Pr\{\overline{X} \leq x_0 + 1/(2n)\}$ replaces $1 - \exp(-b\tilde{u})$ by $2 \sinh(\tilde{u}/2)$.

12.3.3 Approximate conditional inference

To see how saddlepoint approximation may be applied for inference, we consider initially the case where a random sample $Y_1, \dots, Y_n$ from the continuous exponential family $\exp\{\theta y - \kappa(\theta)\}$, $f_0(y)$ depends on a scalar θ . The maximum likelihood estimator $\hat{\theta}$ solves the likelihood equation $\bar{Y} = \kappa'(\hat{\theta})$, so the log likelihood and observed information may be expressed as

$$\ell(\theta; \hat{\theta}) \equiv n \{ \theta \kappa'(\hat{\theta}) - \kappa(\theta) \}, \quad J(\theta; \hat{\theta}) = -\partial^2 \ell(\theta; \hat{\theta}) / \partial \theta^2 = n \kappa''(\theta).$$

Now $\bar{Y}$ is a minimal sufficient statistic for θ , so no information is lost by considering its density or equivalently that of $\hat{\theta}$, which is a 1–1 function of $\bar{Y}$. The cumulant-generating function of Y_j is $K(u) = \kappa(\theta + u) - \kappa(\theta)$, so the saddlepoint equation is $\kappa'(\theta + \tilde{u}) = \bar{y}$. This implies that $\tilde{u} = \hat{\theta} - \theta$; furthermore the second derivative $K''(\tilde{u}) = \kappa''(\theta + \tilde{u}) = n^{-1} J(\hat{\theta}; \hat{\theta})$. Thus the density approximation (12.22) for $\bar{Y}$ is

$$\begin{aligned} f_{\bar{Y}}(\bar{y}; \theta) &\doteq \left\{ \frac{n}{2\pi \kappa''(\tilde{u})} \right\}^{1/2} \exp [n \{ K(\tilde{u}) - \tilde{u} \bar{y} \}] \\ &= \left\{ \frac{n}{2\pi \kappa''(\hat{\theta})} \right\}^{1/2} \exp [n \{ \kappa(\hat{\theta}) - \kappa(\theta) - (\hat{\theta} - \theta) \bar{y} \}]. \end{aligned}$$

Now $\partial \bar{y} / \partial \hat{\theta} = n^{-1} J(\hat{\theta}; \hat{\theta})$, and hence the density of $\hat{\theta}$ may be written as

$$f(\hat{\theta}; \theta) = c |J(\hat{\theta}; \hat{\theta})|^{1/2} \exp \{ \ell(\theta; \hat{\theta}) - \ell(\hat{\theta}; \hat{\theta}) \} \{ 1 + O(n^{-1}) \}, \quad (12.32)$$

where $c = (2\pi)^{-1/2}$. Thus (12.7) again arises as an approximation to the density of a maximum likelihood estimator, here with no ancillary statistic.

Saddlepoint approximation to the cumulative distribution function of $\hat{\theta}$ or equivalently of $\bar{Y}$ gives

$$\Pr(\bar{Y} \leq \bar{y}; \theta) \doteq \Phi \{ r^*(\theta) \}, \quad (12.33)$$

where $r^*(\theta) = r(\theta) + r(\theta)^{-1} \log \{ v(\theta) / r(\theta) \}$, and

$$r(\theta) = \text{sign}(\hat{\theta} - \theta) [2 \{ \ell(\hat{\theta}) - \ell(\theta) \}]^{1/2}, \quad v(\theta) = (\hat{\theta} - \theta) J(\hat{\theta}; \hat{\theta})^{1/2}; \quad (12.34)$$

recall that $\kappa'(\hat{\theta}) = \bar{y}$. An approximate confidence interval for θ may be obtained by finding the values of θ for which $\Phi \{ r^*(\theta) \} = \alpha, 1 - \alpha$ or equivalently for which $r^*(\theta) = z_\alpha, z_{1-\alpha}$. This improves on the likelihood ratio limits, which are found by solving $r(\theta) = z_\alpha, z_{1-\alpha}$.

Example 12.17 (Gamma distribution) Let $Y_1, \dots, Y_n$ be a gamma random sample with mean θ and known shape parameter ν . The log likelihood is then $\ell(\theta; \hat{\theta}) \equiv -n\nu(\log \theta + \hat{\theta}/\theta)$, with $\hat{\theta} = \bar{Y}$ and observed information $J(\theta; \hat{\theta}) = n\nu\theta^{-2}$. Thus the approximate density for $\hat{\theta}$ is

$$c |J(\hat{\theta}; \hat{\theta})|^{1/2} \exp \{ \ell(\theta; \hat{\theta}) - \ell(\hat{\theta}; \hat{\theta}) \} = \left(\frac{n\nu}{2\pi} \right)^{1/2} \frac{1}{\hat{\theta}} \left(\frac{\hat{\theta}}{\theta} \right)^{n\nu} \exp(n\nu - n\nu\hat{\theta}/\theta).$$

Table 12.1 Approximate tail probabilities ($\times 10^2$) corresponding to quantiles of the maximum likelihood estimator of the mean of an exponential variable.

x_p	0.001001	0.01005	0.02532	0.05129	2.996	3.689	4.605	6.908
Exact, $100p$	0.1	1	2.5	5	95	97.5	99	99.9
$100\Phi\{r^*(\theta)\}$	0.104188	1.02937	2.55549	5.08002	94.923	97.454	98.978	99.897
$100\Phi\{r(\theta)\}$	0.029354	0.36040	1.00503	2.21778	90.997	95.189	97.926	99.760
$100\Phi\{v(\theta)\}$	15.88975	16.1099	16.4859	17.1385	97.702	99.642	99.984	100.000

The exact density of $\hat{\theta}$ is gamma with mean θ and shape parameter nv , so the approximation is here exact after renormalization; it merely substitutes Stirling's formula for the gamma function that appears in the exact density.

In this example,

$$r(\theta) = \text{sign}(\hat{\theta} - \theta)[2nv\{\hat{\theta}/\theta - 1 - \log(\hat{\theta}/\theta)\}]^{1/2}, \quad v(\theta) = (nv)^{1/2}(\hat{\theta} - \theta)/\hat{\theta},$$

and Table 12.1 shows that (12.33) is essentially exact for this model when $nv = 1$. The tail probability approximation based on the signed likelihood ratio statistic $r(\theta)$ is also fairly accurate, while that based on the standardized maximum likelihood estimate $v(\theta)$ is very poor. The accuracy of the tail approximation for $r^*(\theta)$ should carry over to the corresponding confidence intervals. ■

Conditional inference

Consider now conditional inference for the scalar parameter ψ in the exponential family

$$f(t_1, t_2; \psi, \lambda) = \exp\{t_1\psi + t_2^T\lambda - \kappa(\psi, \lambda)\} m(t_1, t_2), \quad (12.35)$$

where t_1 is scalar and t_2 has dimension $(p-1) \times 1$, and λ is treated as a $(p-1) \times 1$ nuisance parameter. We obtain an approximate conditional density for T_1 given T_2 by double saddlepoint approximation of

$$f(t_1 | t_2; \psi) = \frac{f(t_1, t_2; \psi, \lambda)}{f(t_2; \psi, \lambda)}. \quad (12.36)$$

The cumulant-generating function of (T_1, T_2) is

$$K(u) = \kappa(\psi + u_1, \lambda + u_2) - \kappa(\psi, \lambda),$$

where $u^T = (u_1, u_2^T)$, u_1 is scalar, and u_2 is a $(p-1) \times 1$ vector. The saddlepoint equation corresponding to the numerator of (12.36) is

$$t_2 = \frac{\partial K(\psi, \lambda + \tilde{u}_2)}{\partial u_2} = \frac{\partial \kappa(\psi, \hat{\lambda}_\psi)}{\partial \lambda} = \kappa_\lambda(\hat{\theta}_\psi),$$

say, where $\hat{\theta}_\psi^T = (\psi, \hat{\lambda}_\psi^T)$ and $\hat{\lambda}_\psi$ is the maximum likelihood estimate of λ with ψ fixed, based on t_2 ; note that $\tilde{u}_2 = \hat{\lambda}_\psi - \lambda$. The matrix of second derivatives

$$\frac{\partial^2 K(\psi, \lambda + \tilde{u}_2)}{\partial u_2 \partial u_2^T} = \frac{\partial^2 \kappa(\psi, \hat{\lambda}_\psi)}{\partial \lambda \partial \lambda^T} = \kappa_{\lambda\lambda}(\hat{\theta}_\psi) = J_{\lambda\lambda}(\hat{\theta}_\psi)$$

For brevity we let κ_λ denote $\partial \kappa / \partial \lambda$, $\kappa_{\psi\lambda}$ denote $\partial^2 \kappa / \partial \psi \partial \lambda^T$, and so forth.

is the observed information for λ when ψ is fixed. The saddlepoint equation corresponding to the denominator of (12.36) is

$$\begin{pmatrix} t_1 \\ t_2 \end{pmatrix} = \begin{pmatrix} \frac{\partial K(\psi + \tilde{u}_1, \lambda + \tilde{u}_2)}{\partial u_1} \\ \frac{\partial K(\psi + \tilde{u}_1, \lambda + \tilde{u}_2)}{\partial u_2} \end{pmatrix} = \begin{pmatrix} \kappa_\psi(\hat{\theta}) \\ \kappa_\lambda(\hat{\theta}) \end{pmatrix},$$

while the matrix of second derivatives is

$$\begin{pmatrix} \frac{\partial^2 K(\psi + \tilde{u}_1, \lambda + \tilde{u}_2)}{\partial u_1^2} & \frac{\partial^2 K(\psi + \tilde{u}_1, \lambda + \tilde{u}_2)}{\partial u_1 \partial u_2} \\ \frac{\partial^2 K(\psi + \tilde{u}_1, \lambda + \tilde{u}_2)}{\partial u_1 \partial u_2} & \frac{\partial^2 K(\psi + \tilde{u}_1, \lambda + \tilde{u}_2)}{\partial u_2^2} \end{pmatrix} = \begin{pmatrix} \kappa_{\psi\psi}(\hat{\theta}) & \kappa_{\psi\lambda}(\hat{\theta}) \\ \kappa_{\lambda\psi}(\hat{\theta}) & \kappa_{\lambda\lambda}(\hat{\theta}) \end{pmatrix} = J(\hat{\theta}),$$

where $\hat{\theta}^T = (\hat{\psi}, \hat{\lambda}^T)$ is the overall maximum likelihood estimate. Substitution of these formulae into the double saddlepoint approximation (12.26) and then reorganization along the lines that leads to (12.32) gives

$$f(t_1 | t_2; \psi) \doteq \left\{ \frac{|J_{\lambda\lambda}(\hat{\theta}_\psi)|}{2\pi |J(\hat{\theta})|} \right\}^{1/2} \exp \{ \ell(\hat{\theta}_\psi) - \ell(\hat{\theta}) \}, \quad (12.37)$$

where $\ell(\theta) = \ell(\psi, \lambda) = \log f(t_1, t_2; \psi, \lambda)$ is the overall log likelihood.

Approximate tail probabilities associated with particular values of ψ can be formed in the way outlined after (12.33), and are used to construct confidence intervals. The basis of the tail probability approximation is (12.27), which here depends on ψ , and is

$$\Pr(T_1 \leq t_1 | T_2 = t_2; \psi) \doteq \Phi \{ r^*(\psi) \}, \quad (12.38)$$

where $r^*(\psi) = r(\psi) + r(\psi)^{-1} \log \{ r(\psi) / v(\psi) \}$, with

$$r(\psi) = \text{sign}(\hat{\psi} - \psi) [2\{\ell(\hat{\theta}) - \ell(\hat{\theta}_\psi)\}]^{1/2}, \quad v(\psi) = (\hat{\psi} - \psi) \left\{ \frac{|J(\hat{\theta})|}{|J_{\lambda\lambda}(\hat{\theta}_\psi)|} \right\}^{1/2}.$$

Thus the improved approximation again involves modifying the signed likelihood ratio statistic, here using a standardized maximum likelihood estimate rather than the score statistic that appeared with the regression-scale model. We can write

$$v(\psi) = (\hat{\psi} - \psi) \left\{ \frac{|J(\hat{\theta}_\psi)|}{|J_{\lambda\lambda}(\hat{\theta}_\psi)|} \right\}^{1/2} \times \left\{ \frac{|J(\hat{\theta})|}{|J(\hat{\theta}_\psi)|} \right\}^{1/2} = \gamma \times C, \quad (12.39)$$

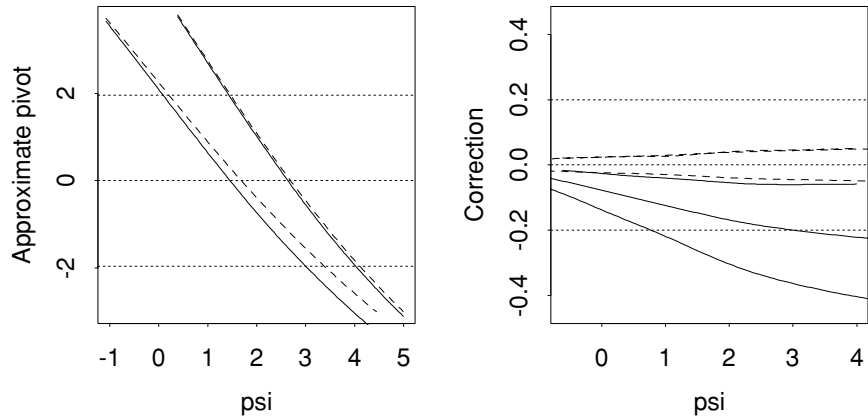
say, yielding a three-part decomposition of $r^*(\psi)$ like (12.18). It is an exercise to show that $r^*(\psi)$ is invariant to interest-preserving reparametrization.

Example 12.18 (Nodal involvement data) A central issue for the data in Table 10.8 is how nodal involvement depends on the five binary covariates. For purpose of illustration we consider setting confidence intervals for the parameter ψ associated with *acid*, and regard the parameters for other covariates as incidental. The left panel of Figure 12.8 shows $r(\psi)$ and $r^*(\psi)$ as functions of ψ , when *acid* only is included, and when all five covariates are included. The small-sample modification is appreciably larger when several nuisance parameters are eliminated, and the standard error, corresponding roughly to the slope, is larger. The right panel shows information and nuisance parameter corrections r_{INF} and r_{NP} for models with *1+acid*, *1+stage+xray+acid*, and with all five covariates. All three information corrections

Table 12.2 Estimate, standard errors, and 0.95 confidence intervals for the coefficient of *acid*, ψ , for the nodal involvement data with all the other covariates fitted. The continuity corrected version of $r^*(\psi)$ is obtained by multiplying $v(\psi)$ by $(e^{\psi-\psi} - 1)/(\hat{\psi} - \psi)$.

Method	Estimate (SE)	Interval
Normal approximation to $\hat{\psi}$	1.68 (0.79)	(0.136, 3.232)
Normal approximation to $\hat{\psi}_a$	1.49 (0.74)	(0.048, 2.930)
Directed deviance $r(\psi)$	—	(0.209, 3.378)
Modified directed deviance $r^*(\psi)$	—	(0.086, 2.998)
Modified directed deviance $r^*(\psi)$ with continuity correction	—	(−0.131, 3.330)

Figure 12.8 Conditional inference for coefficient of *acid*, ψ , for nodal involvement data. Left: signed likelihood ratio statistic $r(\psi)$ (dashes) and modified version $r^*(\psi)$ (solid) for models with five nuisance parameters (left pair of curves) and with one nuisance parameter (right pair of curves, shifted right by one unit). The dotted horizontal lines are at $0, \pm 1.96$. Right: information corrections r_{INF} (dashes) and nuisance parameter corrections r_{NP} (solid) for models with one, three, and five nuisance parameters. The r_{NP} increase in size with the number of nuisance parameters eliminated. The dotted horizontal lines at ± 0.2 show a rule of thumb for substantial corrections.



are small, but the nuisance parameter correction increases sharply as more parameters are eliminated.

An approximate conditional maximum likelihood estimate $\hat{\psi}_a$ is obtained as the solution to $r^*(\psi) = 0$, with standard error given by the slope of $r^*(\psi)$ at that point. Together these yield a confidence interval for ψ based on normal approximation. Table 12.2 shows that $\hat{\psi}_a$ has smaller magnitude and standard error than has the ordinary maximum likelihood estimate $\hat{\psi}$. Small-sample adjustment shortens confidence intervals and moves them closer to zero. In such cases a continuity correction can be applied to give results closer to those from exact conditioning. This gives appreciably wider confidence intervals, as the discussion in Example 7.38 would suggest. ■

Curved exponential family

Our previous discussion has applied to linear exponential families, but curved exponential family models are also important in applications. The key difficulty in developing small-sample inferences is then to find an analogue of the quantities (12.17) and (12.39) that modify the signed likelihood ratio statistic. An exact expression is unavailable but one of several possible approximations is

$$v(\psi) \doteq |I(\hat{\theta})|^{-1} |C(\hat{\theta}, \hat{\theta}_\psi)| \{ |J(\hat{\theta})| / |J_{\lambda\lambda}(\hat{\theta}_\psi)| \}^{1/2}, \quad (12.40)$$

with $I(\theta)$ the expected information and $C(\theta, \theta_0)$ denoting the $p \times p$ matrix

$$\begin{pmatrix} \text{cov}_0 \{ \ell(\theta_0) - \ell(\theta), \ell_\theta(\theta_0) \} \\ \text{cov}_0 \{ \ell_\lambda(\theta), \ell_\theta(\theta_0) \} \end{pmatrix}.$$

Here $\ell_\theta(\theta) = \partial \ell(\theta) / \partial \theta$, and so forth, and cov_0 denotes covariance taken with respect to the model with parameter value θ_0 . Expression (12.40) admits a decomposition into two parts like that at (12.39). It requires neither that an ancillary statistic be specified nor does it involve sample space derivatives, but it does entail a loss of precision relative to (12.39), use of which in continuous models gives relative error of sizes $O(n^{-3/2})$ in moderate deviation regions and $O(n^{-1/2})$ in large deviation regions. By contrast use of (12.40) reduces the relative error in moderate deviation regions to $O(n^{-1})$. The key point, however, is that the relative error properties, which give highly accurate approximations far into the tails of the distribution of the modified signed likelihood ratio statistic $R^*(\psi)$, are preserved.

Example 12.19 (Nonlinear model) Consider a model in which the $n \times 1$ vector of responses $Y \sim N_n(\eta, \sigma^2 I_n)$ and the $n \times 1$ mean vector $\eta = \eta(\beta)$ depends on a $p \times 1$ parameter vector β . Let the parameter of interest ψ be an element of β , β_1 , say, let λ denote the $p \times 1$ vector comprising the remaining elements of β and σ^2 , and let $\theta^T = (\psi, \lambda^T)$. The log likelihood,

$$\ell(\theta) \equiv -\frac{1}{2} \{n \log \sigma^2 + (y - \eta)^T (y - \eta) / \sigma^2\},$$

has score vector given by

$$\ell_\theta(\theta)^T = (\sigma^{-2} (y - \eta)^T \eta_\beta, \{(y - \eta)^T (y - \eta) - n\sigma^2\} / (2\sigma^4))$$

where η_β is the $n \times p$ matrix $\partial \eta / \partial \beta^T$. On using the decomposition

$$\begin{aligned} \ell(\theta) &= -\frac{1}{2} \{n \log \sigma^2 + (y - \eta_0 + \eta_0 - \eta)^T (y - \eta_0 + \eta_0 - \eta) / \sigma^2\} \\ &= -\frac{1}{2} \{(y - \eta_0)^T (y - \eta_0) + 2(y - \eta_0)^T (\eta_0 - \eta)\} / \sigma^2 + d, \end{aligned}$$

where d depends only on the parameters and subscript 0 indicates that a quantity is evaluated at $\theta_0 = (\beta_0, \sigma_0^2)$, we find

$$\text{cov}_0 \{\ell(\theta), \ell_\theta(\theta_0)\} = -\sigma^{-2} \{(\eta_0 - \eta)^T \eta_{\beta 0}, n/2\},$$

from which we see that the first row of $C(\theta, \theta_0)$ equals

$$((\eta_0 - \eta)^T \eta_{\beta 0} / \sigma^2, n(\sigma^{-2} - \sigma_0^{-2}) / 2),$$

and a similar calculation shows that the remaining $p \times (p + 1)$ submatrix is

$$\text{cov}_0 \{\ell_\lambda(\theta), \ell_\theta(\theta_0)\} = \begin{pmatrix} \sigma^{-2} \eta_{\beta 2}^T \eta_{\beta 0} & 0 \\ \sigma^{-4} (\eta_0 - \eta)^T \eta_{\beta 0} & n / (2\sigma^4) \end{pmatrix}.$$

Thus the approximation to $v(\psi)$ involves this matrix, evaluated with $\theta_0 = \hat{\theta}$ and $\theta = \hat{\theta}_\psi$, and the other information matrices. In this case $I(\hat{\theta}) = J(\hat{\theta})$. ■

Example 12.20 (Calcium data) For numerical illustration we consider the data of Example 10.1, to which we fit a normal model with constant variance σ^2 and mean $\beta_0 \{1 - \exp(-x/\beta_1)\}$, where x represents time in minutes. First-order inference was discussed in Example 10.9.

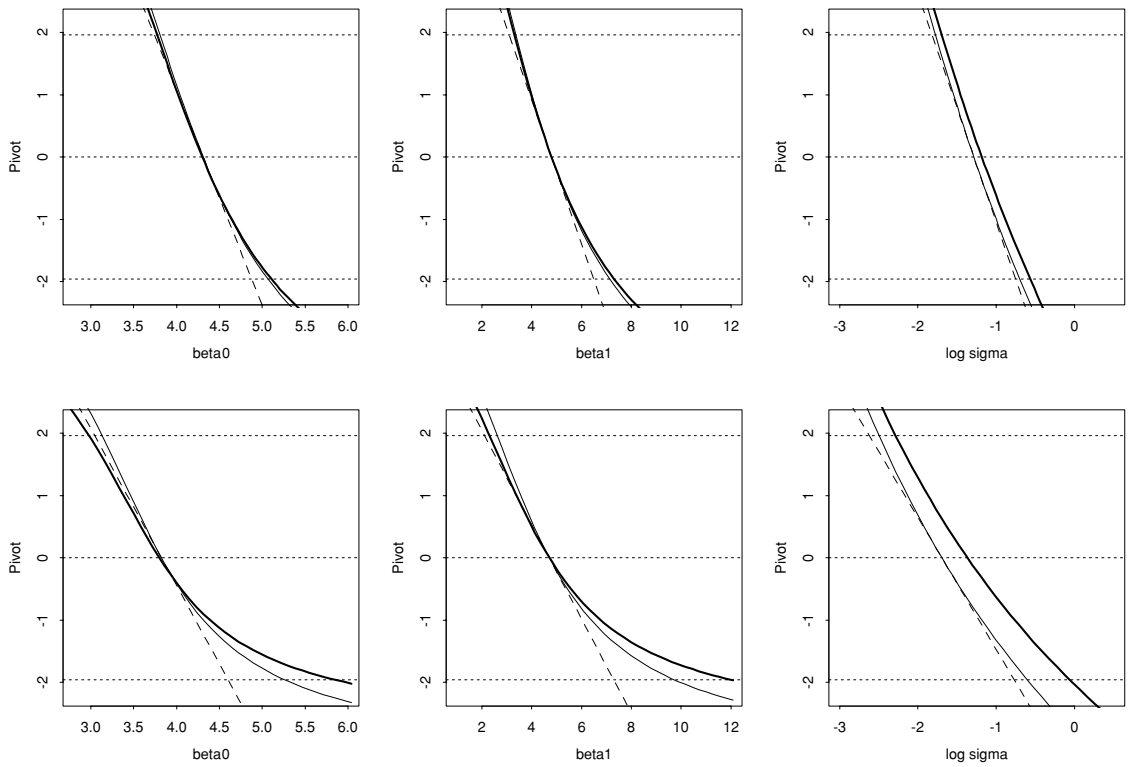


Figure 12.9 Conditional inference for parameters of nonlinear regression model for calcium data. Each panel shows how the signed likelihood ratio statistic (solid), the modified signed likelihood ratio statistic (heavy), and the standardized maximum likelihood estimate (dashes) depend on the corresponding parameter. The dotted horizontal lines are at $0, \pm 1.96$. Upper panels: full data; lower panels: just one observation at each time.

The result of applying the computations in Example 12.19 are given in Figure 12.9, which shows, for each parameter of interest ψ , the quantities $r(\psi)$, $r^*(\psi)$ and the pivot $(\hat{\psi} - \psi)/s_\psi$ from normal approximation to the maximum likelihood estimator; s_ψ is the standard error for $\hat{\psi}$ based on the observed information matrix. Results are given for the full dataset, and for a reduced dataset comprising just the nine responses in the final column of Table 10.1. The shallower slopes in the lower panels show the effect of reducing the sample size.

Normal approximation is clearly inadequate, particularly at the upper limits of confidence intervals, but the ordinary unmodified signed likelihood ratio statistic yields intervals for β_0 and β_1 that are reasonably close to those from the modified version, at least for the full dataset. The unmodified estimates of $\log \sigma$ are strongly biased downwards, and the higher-order procedures give worthwhile improvements. Much of this bias may be removed by dividing the residual sum of squares by denominator $n - p$ rather than n .

It seems wise to use the modified statistics for all three parameters with the smaller dataset. The 0.95 confidence intervals based on $r(\beta_1)$ and $r^*(\beta_1)$ are (2.63, 9.75) and (2.31, 12.02), for instance, while the corresponding intervals for σ are (0.08, 0.55) and (0.10, 0.94), whose right tails differ substantially. ■

Variants have been proposed that replace the expectations by averages in order to give a purely empirical version of (12.40); see the bibliographic notes.

Exercises 12.3

- Let Y and X be independent exponential variables with means $1/(\lambda + \psi)$ and $1/\lambda$. Find the distribution of Y given $X + Y$ and show that when $\psi = 0$ it has mean $s/2$ and variance $s^2/12$. Construct an exact conditional test of the hypothesis $E(Y) = E(X)$.
- A discrete exponential family for independent pairs $(S_1, T_1), \dots, (S_n, T_n)$ has form

$$\exp \{s_j \lambda_j + t_j \psi_j + \kappa(\lambda_j, \psi_j)\} c_j(s_j, t_j), \quad j = 1, \dots, n.$$

It is intended to test the hypothesis $\psi_1 = \dots = \psi_n$ regardless of the values of the λ_j . Generalize Example 12.14 to explain how to perform Monte Carlo exact significance tests. Give the acceptance probability for the Metropolis–Hastings algorithm, and write out the algorithm when

$$c_j(s_j, t_j) = \{(s_j - t_j)! t_j!\}^{-1}, \quad s_j = t_j, t_j + 1, \dots, \quad t_j = 0, 1, \dots$$

How does the argument change if the variables are continuous?

- Show that the saddlepoint approximations for the density and distribution of the average of a normal random sample are exact.
- Show that saddlepoint approximation for the inverse Gaussian density

$$f(y; \mu, \lambda) = \left(\frac{\lambda}{2\pi y^3} \right)^{1/2} \exp \{ -\lambda(y - \mu)^2 / (2\mu^2 y) \}, \quad y > 0, \lambda, \mu > 0,$$

is exact after renormalization. Investigate the accuracy of the distribution function approximation (12.23) when $n = 1$.

- Consider independent exponential observations Y_j with means $(x_j^T \beta)^{-1}$, for $j = 1, \dots, n$, where the x_j are $p \times 1$ vectors of explanatory variables and the $p \times 1$ parameter vector β is unknown. Find the ingredients of (12.37) and (12.38) when inference is required for $\psi = \beta_1$, with $\lambda = (\beta_2, \dots, \beta_p)$ treated as incidental. Give the exact conditional likelihood for ψ when $Y_1, \dots, Y_{n-1}$ have mean λ^{-1} and Y_n has mean $(\lambda + \psi)^{-1}$, and investigate the accuracy of (12.38) when $n = 2$.
- Show that (12.40) retrieves (12.39) for a linear exponential family (12.35).
- Check the details of Examples 12.19 and 12.20. Find (12.40) when σ^2 is of interest.

12.4 Modified Profile Likelihood

12.4.1 Likelihood adjustment

The profile log likelihood is a standard tool for inference in large-sample situations, and it is natural to consider if and how it may be modified for use in small-sample problems. We saw at (12.13), for instance, that a log marginal likelihood for parameters ψ controlling the variance of a normal distribution could be obtained by adding a term to the profile log likelihood, while (12.37) suggests that an approximate conditional likelihood for a linear exponential family model has form

$$\ell(\psi, \hat{\lambda}_\psi) + \frac{1}{2} \log |J_{\lambda\lambda}(\psi, \hat{\lambda}_\psi)|,$$

where $\widehat{\lambda}_\psi$ is the maximum likelihood estimate of λ for fixed ψ and $J_{\lambda\lambda}(\psi, \lambda)$ is the corner of the observed information matrix corresponding to λ . This amounts to a penalization of the log likelihood by an amount that depends on the information available for λ ; when this is large, the profile log likelihood is more strongly penalized than when it is small.

The form of these expressions suggests that in general settings we multiply the profile likelihood

$$L_p(\psi) = \exp \{ \ell_p(\psi) \} = \exp \{ \ell(\psi, \widehat{\lambda}_\psi) \}$$

by a cunningly chosen function of ψ , giving a *modified profile likelihood*

$$L_{mp}(\psi) = \exp \{ \ell_{mp}(\psi) \} = M(\psi) L_p(\psi). \quad (12.41)$$

It is natural to try and choose $M(\psi)$ so that $L_{mp}(\psi)$ gives inferences equivalent to using a marginal or conditional likelihood for ψ , if such is available. It is a remarkable fact that taking

$$M(\psi) = |J_{\lambda\lambda}(\psi, \widehat{\lambda}_\psi)|^{-1/2} \left| \frac{\partial \widehat{\lambda}}{\partial \widehat{\lambda}_\psi^\top} \right| \quad (12.42)$$

achieves this to a high degree of accuracy in some generality. The first term of (12.42) involves the part of the observed information matrix mentioned above, while the second term is a Jacobian needed if the modified profile likelihood is to be invariant to interest-preserving transformations: typically this term depends on the parameter ψ . A derivation of (12.42) is given after Example 12.18. Here is a toy example.

Example 12.21 (Normal linear model) Let the parameter of interest in the normal linear model be the variance σ^2 , with the $p \times 1$ vector β treated as incidental. The log likelihood is

$$\ell(\beta, \sigma^2) \equiv -\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (y - X\beta)^\top (y - X\beta),$$

and the maximum likelihood estimator of β for σ^2 fixed is $\widehat{\beta}_{\sigma^2} = (X^\top X)^{-1} X^\top y$. As this is independent of σ^2 , we have $\widehat{\beta} = \widehat{\beta}_{\sigma^2}$. Thus

$$J_{\beta\beta}(\beta, \sigma^2) = \sigma^{-2} X^\top X, \quad \frac{\partial \widehat{\beta}_{\sigma^2}^\top}{\partial \widehat{\beta}} = I_p, \quad M(\sigma^2) = (\sigma^2)^{p/2} |X^\top X|^{-1/2},$$

and the modified profile log likelihood is

$$\begin{aligned} \ell_{mp}(\sigma^2) &\equiv -\frac{n-p}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (y - X\widehat{\beta})^\top (y - X\widehat{\beta}) \\ &= -\frac{n-p}{2} (\log \sigma^2 - S^2/\sigma^2), \end{aligned}$$

where S^2 is the unbiased estimator of σ^2 . Hence (12.41) produces inferences identical to those based on the marginal distribution of the residual sum of squares, or equivalently the marginal likelihood for σ^2 ; see Example 12.9.

In fact $|\partial \widehat{\beta}_{\sigma^2}^\top / \partial \widehat{\beta}| = 1$ in all regression-scale models (Exercise 12.4.2). ■

Computation of (12.41) demands knowledge of the two terms of (12.42). In applications the first is easily found by numerical or analytical differentiation of $\ell(\psi, \lambda)$, but the second is a sample space derivative and approximations to it are usually needed. When the log likelihood can be written in terms of the maximum likelihood estimates $\hat{\psi}$ and $\hat{\lambda}$ and an ancillary statistic a , however, the equation determining $\hat{\lambda}_\psi$ may be expressed as

$$\frac{\partial \ell(\psi, \hat{\lambda}_\psi; \hat{\psi}, \hat{\lambda}, a)}{\partial \lambda} = 0,$$

and partial differentiation with respect to $\hat{\lambda}$, holding ψ , $\hat{\psi}$, and a fixed, yields

$$\frac{\partial^2 \ell(\psi, \hat{\lambda}_\psi; \hat{\psi}, \hat{\lambda}, a)}{\partial \lambda \partial \lambda^T} \frac{\partial \hat{\lambda}_\psi^T}{\partial \hat{\lambda}} + \frac{\partial^2 \ell(\psi, \hat{\lambda}_\psi; \hat{\psi}, \hat{\lambda}, a)}{\partial \lambda \partial \hat{\lambda}^T} = 0.$$

This gives the alternative and often more convenient expression

$$\left| \frac{\partial \hat{\lambda}_\psi^T}{\partial \hat{\lambda}} \right| = |J_{\lambda\lambda}(\psi, \hat{\lambda}_\psi; \hat{\psi}, \hat{\lambda}, a)|^{-1} \left| \frac{\partial^2 \ell(\psi, \hat{\lambda}_\psi; \hat{\psi}, \hat{\lambda}, a)}{\partial \lambda \partial \hat{\lambda}^T} \right|. \quad (12.43)$$

Although the ancillary must be held fixed when differentiating with respect to $\hat{\lambda}$, it is not needed explicitly, and in some cases (12.43) can be obtained without the burden of specifying a .

Example 12.22 (Linear exponential family) In a linear exponential family with log likelihood expressed as

$$\ell(\psi, \lambda) \equiv t_1^T \psi + t_2^T \lambda - \kappa(\psi, \lambda)$$

there is no ancillary statistic and the maximum likelihood estimates $\hat{\psi}$ and $\hat{\lambda}$ are solutions of the equations

$$t_1 = \kappa_\psi(\hat{\psi}, \hat{\lambda}), \quad t_2 = \kappa_\lambda(\hat{\psi}, \hat{\lambda}).$$

Thus the log likelihood may be written

$$\ell(\psi, \lambda; \hat{\psi}, \hat{\lambda}) = \kappa_\psi(\hat{\psi}, \hat{\lambda})^T \psi + \kappa_\lambda(\hat{\psi}, \hat{\lambda})^T \lambda - \kappa(\psi, \lambda),$$

and (12.43) implies that

$$\left| \frac{\partial \hat{\lambda}_\psi^T}{\partial \hat{\lambda}} \right| = |\kappa_{\lambda\lambda}(\psi, \hat{\lambda}_\psi)|^{-1} |\kappa_{\lambda\lambda}(\hat{\psi}, \hat{\lambda})| = |J_{\lambda\lambda}(\psi, \hat{\lambda}_\psi; \hat{\psi}, \hat{\lambda})|^{-1} |J_{\lambda\lambda}(\hat{\psi}, \hat{\lambda}; \hat{\psi}, \hat{\lambda})|.$$

Thus the modified profile log likelihood for ψ is

$$\ell_{\text{mp}}(\psi) \equiv \ell_p(\psi) + \frac{1}{2} \log |J_{\lambda\lambda}(\psi, \hat{\lambda}_\psi; \hat{\psi}, \hat{\lambda})|,$$

where terms independent of ψ have been neglected. Hence (12.41) and (12.42) do indeed retrieve the approximate conditional likelihood (12.37), apart from constants of proportionality. ■

Inference on ψ is performed by treating (12.41) as a likelihood. Its maximizing value $\hat{\psi}_{\text{mp}}$ yields a confidence interval for ψ , for example by normal approximation

with variance based on the second derivative of $\ell_{\text{mp}}(\psi)$ at $\hat{\psi}_{\text{mp}}$, or by chi-squared approximation to the distribution of $2\{\ell_{\text{mp}}(\hat{\psi}_{\text{mp}}) - \ell_{\text{mp}}(\psi)\}$. The maximum likelihood estimator based on L_{mp} typically has better properties than $\hat{\psi}$, as is the case in Example 12.21, but likelihood modification is not a universal panacea. One difficulty is that the underlying inferential basis is first-order distributional results such as normal approximation to the distribution of the maximum modified likelihood estimator rather than more accurate forms such as (12.38). Another difficulty is that modification may not be enough to remove inconsistency of maximum likelihood estimators.

Example 12.23 (Binary matched pairs) Consider matched pairs of binary observations R_{0j}, R_{1j} , with success probabilities

$$\pi_{0j} = \frac{e^{\lambda_j}}{1 + e^{\lambda_j}}, \quad \pi_{1j} = \frac{e^{\lambda_j + \psi}}{1 + e^{\lambda_j + \psi}}, \quad j = 1, \dots, n.$$

This logistic regression model is a linear exponential family with minimal sufficient statistic $S_1, \dots, S_n, T$, where $S_j = R_{0j} + R_{1j}$ is associated with λ_j and $T = \sum R_{1j}$ with ψ . We suppose that ψ is finite, and compare its maximum likelihood estimators based on the conditional, modified profile, and usual likelihoods as $n \rightarrow \infty$.

Pairs for which $S_j = 0$ or 2 are uninformative (Problem 12.11), and we suppose that they have already been dropped from the analysis, so all n pairs are discordant, that is, $S_1 = \dots = S_n = 1$.

The exact conditional likelihood for ψ is obtained from the conditional density of $T = \sum R_{1j}$ given that $S_1 = \dots = S_n = 1$. Conditional on $S_j = 1$, R_{1j} is a Bernoulli variable with success probability $\pi = e^{\psi}/(1 + e^{\psi})$, so T has a binomial distribution with denominator n and probability π . Thus the conditional log likelihood is $\ell_c(\psi) \equiv T\psi - n \log(1 + e^{\psi})$, which is maximized at $\hat{\psi}_c = \log\{T/(n - T)\} \xrightarrow{P} \psi$ as $n \rightarrow \infty$; hence $\hat{\psi}_c$ is consistent.

The overall log likelihood is

$$\ell(\psi, \lambda) \equiv T\psi + \sum_{j=1}^n \{\lambda_j - \log(1 + e^{\lambda_j}) - \log(1 + e^{\psi + \lambda_j})\},$$

and the values of λ_j that maximise $\ell(\psi, \lambda)$ for fixed ψ all equal $\hat{\lambda}_{\psi} = -\frac{1}{2}\psi$. Thus the profile log likelihood is

$$\ell_p(\psi) = T\psi - 2n \log(1 + e^{\psi/2}).$$

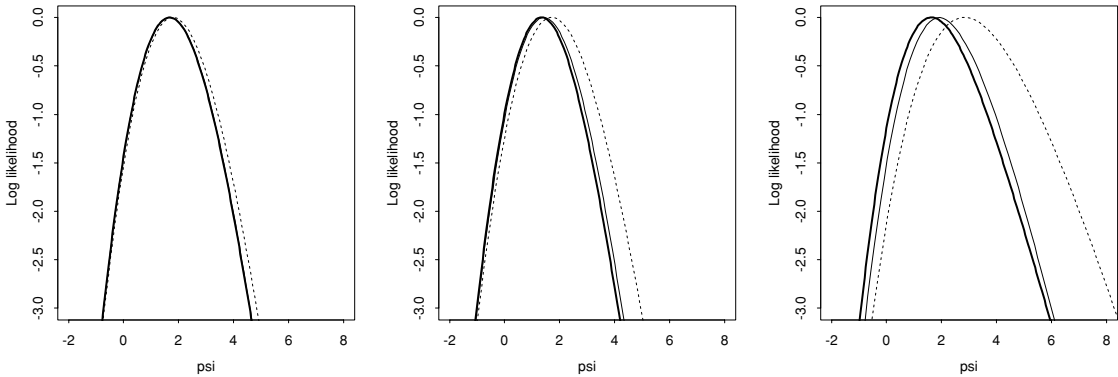
It is straightforward to see that the maximum likelihood estimator $\hat{\psi} \xrightarrow{P} 2\psi$ as $n \rightarrow \infty$ (Problem 12.11). Thus $\hat{\psi}$ is inconsistent.

Differentiation of $\ell(\psi, \lambda)$ establishes that $J_{\lambda\lambda}(\psi, \hat{\lambda}_{\psi})$ is a $n \times n$ matrix with elements $2\gamma/(1 + \gamma)^2$ on the diagonal and zeros elsewhere, where $\gamma = \exp(\psi/2)$. Hence the modified profile log likelihood is

$$\ell_{\text{mp}}(\psi) \equiv \frac{1}{4}(n + 4T)\psi - 3n \log(1 + e^{\psi/2}),$$

ψ	0	0.5	1	1.5	2	2.5	3	4	5
Limit of $\hat{\psi}_c$	0	0.5	1	1.5	2	2.5	3	4	5
Limit of $\hat{\psi}_{mp}$	0	0.66	1.27	1.81	2.23	2.56	2.79	3.05	3.16
Limit of $\hat{\psi}$	0	1	2	3	4	5	6	8	10

Table 12.3 Probability limits of ordinary, conditional, and maximum modified likelihood estimators of log odds ratio ψ in binary matched pairs.



which is maximized at $\hat{\psi}_{mp} = 2 \log\{(1 + 4T/n)/(5 - 4T/n)\}$. Now $T/n \xrightarrow{P} \pi$ as $n \rightarrow \infty$, so $\hat{\psi}_{mp} \xrightarrow{P} 2 \log\{(1 + 5e^\psi)/(5 + e^\psi)\}$.

Table 12.3 compares these limiting values. Although $\hat{\psi}_{mp}$ is inconsistent, it is not horribly biased in the range $|\psi| < 3$ usually met in applications: the modification is partly but not wholly successful in eliminating bias. ■

Example 12.24 (Nodal involvement data) For numerical illustration of likelihood modification we fit logistic regression models to the rows of Table 10.8 with $m = 2$ or $m = 1$: this gives 17 binary responses in all. We let ψ be the parameter corresponding to *acid*, and fit models with terms *1+acid*, *1+stage+xray+acid*, and with all five covariates; thus the nuisance parameter λ has dimensions $q = 1, 3$, and 5 .

Figure 12.10 shows the conditional, modified profile, and profile log likelihoods for these three models. The conditional likelihoods were obtained by symbolic computation of the necessary generating functions; see Example 12.13. The profile log likelihoods are quite different from the conditional and modified profile log likelihoods. The effect of modification depends on the number of nuisance parameters; it works almost perfectly when $q = 1$ but less well otherwise. In the right panel five nuisance parameters are eliminated from a likelihood based on just 17 binary observations, and it is perhaps surprising that modification works so well.

Table 12.4 compares the corresponding estimates and confidence intervals. The difference between results based on the profile likelihood and conditional and modified profile likelihoods increases sharply with the number of nuisance parameters, but the difference between these two last sets of results remains modest. ■

Figure 12.10 Likelihood inference for nodal involvement data. Left panel: conditional (heavy), modified profile (solid) and profile log likelihoods (dots) for model with terms *1+acid*. Centre and right panels: corresponding log likelihoods with terms *1+stage+xray+acid* and with all five covariates.

Table 12.4 Estimates, standard errors, and 0.95 confidence intervals based on profile, modified profile and conditional likelihoods for reduced nodal data having 17 binary responses, with $q = 1, 3$ and 5 nuisance parameters.

q	Estimate (SE)		Confidence interval		
	Profile	Modified	Profile	Modified	Conditional
1	1.79 (1.08)	1.68 (1.04)	(−0.21, 4.15)	(−0.28, 3.86)	(−0.26, 3.92)
3	1.72 (1.13)	1.43 (1.02)	(−0.41, 4.21)	(−0.54, 3.51)	(−0.53, 3.48)
5	2.83 (1.61)	1.91 (1.15)	(0.13, 6.90)	(−0.39, 4.57)	(−0.44, 4.71)

Derivation of (12.42)

Consider a model with interest and nuisance parameters ψ and λ , and for which the data $y \mapsto (\hat{\psi}, \hat{\lambda}, a)$, where a is ancillary. If a factorization

$$f(\hat{\psi}, \hat{\lambda} \mid a; \psi, \lambda) = f(\hat{\psi} \mid a; \psi) f(\hat{\lambda} \mid \hat{\psi}, a; \psi, \lambda) \quad (12.44)$$

holds, then a marginal likelihood for ψ may be based on the first term on the right. We now apply the p^* formula to the left-hand density, giving

$$(2\pi)^{-p/2} \bar{c}_1(\psi, \lambda, a) |J(\hat{\psi}, \hat{\lambda})|^{1/2} \exp \{ \ell(\psi, \lambda) - \ell(\hat{\psi}, \hat{\lambda}) \}, \quad (12.45)$$

where (ψ, λ) has dimension p and $\bar{c}_1(\psi, \lambda, a) = 1 + O_p(n^{-1})$. To approximate to $f(\hat{\lambda} \mid \hat{\psi}, a; \psi, \lambda)$, we note that with $\psi, \hat{\psi}$, and a held fixed we have

$$f(\hat{\lambda} \mid \hat{\psi}, a; \psi, \lambda) = f(\hat{\lambda}_\psi \mid \hat{\psi}, a; \psi, \lambda) \left| \frac{\partial \hat{\lambda}_\psi}{\partial \hat{\lambda}} \right|,$$

and apply the p^* formula to the density on the right, giving

$$(2\pi)^{-q/2} \bar{c}_2(\psi, \lambda, a) |J_{\lambda\lambda}(\psi, \hat{\lambda}_\psi)|^{1/2} \exp \{ \ell(\psi, \lambda) - \ell(\psi, \hat{\lambda}_\psi) \} \left| \frac{\partial \hat{\lambda}_\psi}{\partial \hat{\lambda}} \right|, \quad (12.46)$$

where q is the dimension of λ and $\bar{c}_2(\psi, \lambda, a) = 1 + O_p(n^{-1})$. On substituting (12.45) and (12.46) into (12.44) and rearranging we find that $f(\hat{\psi} \mid a; \psi)$ equals

$$(2\pi)^{-(p-q)/2} \bar{c}(\psi, \lambda, a) \frac{|J(\hat{\psi}, \hat{\lambda})|^{1/2}}{|J_{\lambda\lambda}(\psi, \hat{\lambda}_\psi)|^{1/2}} \exp \{ \ell(\psi, \hat{\lambda}_\psi) - \ell(\hat{\psi}, \hat{\lambda}) \} \left| \frac{\partial \hat{\lambda}}{\partial \hat{\lambda}_\psi} \right|,$$

where $\bar{c}(\psi, \lambda, a) = \bar{c}_1(\psi, \lambda, a) / \bar{c}_2(\psi, \lambda, a) = 1 + O(n^{-1})$. Thus

$$f(\hat{\psi} \mid a; \psi) \propto \exp \{ \ell_p(\psi) \} M(\psi) \{ 1 + O(n^{-1}) \}, \quad (12.47)$$

as a function of ψ , with $M(\psi)$ given by (12.42). It can be shown that the error term is $O(n^{-3/2})$ in a moderate deviation region, that is, when $\hat{\psi}$ differs from the true ψ by only $O(n^{-1/2})$.

Exercise 12.4.4 derives (12.47) as an approximate conditional likelihood.

12.4.2 Parameter orthogonality

In applications it is rarely possible to find an explicit expression for the second term of (12.42). Approximations to it are available in certain cases, but when they are not,

This section demands a nodding acquaintance with partial differential equations.

it is natural to seek to reduce the importance of that term. If $\widehat{\lambda}_\psi$ is independent of ψ , then $\widehat{\lambda}_\psi = \widehat{\lambda}$ for all ψ , and $|\partial\widehat{\lambda}/\partial\lambda_\psi|$ plays no part in the likelihood modification. Although this occurs only in particular models, it suggests that we seek to reduce the dependence of $\widehat{\lambda}_\psi$ on ψ more generally. One approach to this is through orthogonal parameters.

To motivate subsequent discussion, let $\bar{\ell} = n^{-1}\ell$ denote the log likelihood, standardized to be of order one, and note that $\widehat{\lambda}_\psi$ is determined by the equation $\bar{\ell}_\lambda(\psi, \widehat{\lambda}_\psi) = 0$. If we suppose that $\widehat{\psi} - \psi = O(n^{-1/2})$, then Taylor series expansion around the overall maximum likelihood estimator gives

$$0 = \bar{\ell}_\lambda(\widehat{\psi}, \widehat{\lambda}) + \bar{\ell}_{\lambda\psi}(\widehat{\psi}, \widehat{\lambda})(\psi - \widehat{\psi}) + \bar{\ell}_{\lambda\lambda}(\widehat{\psi}, \widehat{\lambda})(\widehat{\lambda}_\psi - \widehat{\lambda}) + O_p(n^{-1}),$$

where second and higher derivatives such as $\bar{\ell}_{\lambda\psi}(\widehat{\psi}, \widehat{\lambda})$ are $O_p(1)$. Hence

$$\begin{aligned}\widehat{\lambda}_\psi - \widehat{\lambda} &= \bar{\ell}_{\lambda\lambda}(\widehat{\psi}, \widehat{\lambda})^{-1} \bar{\ell}_{\lambda\psi}(\widehat{\psi}, \widehat{\lambda})(\widehat{\psi} - \psi) + O_p(n^{-1}) \\ &= J_{\lambda\lambda}(\widehat{\psi}, \widehat{\lambda})^{-1} J_{\lambda\psi}(\widehat{\psi}, \widehat{\lambda})(\widehat{\psi} - \psi) + O_p(n^{-1}) \\ &= I_{\lambda\lambda}(\psi, \lambda)^{-1} I_{\lambda\psi}(\psi, \lambda)(\widehat{\psi} - \psi) + O_p(n^{-1}),\end{aligned}$$

where $I_{\lambda\lambda}$ and $I_{\lambda\psi}$ are components of the expected information matrix. In regular models these and the corresponding observed information quantities are of order n , so $\widehat{\lambda}_\psi - \widehat{\lambda}$ will be of precise order $n^{-1/2}$ unless the model is set up so that $J_{\lambda\psi}(\widehat{\psi}, \widehat{\lambda})$ or $I_{\lambda\psi}(\psi, \lambda)$ vanishes. If this can be arranged, then $\widehat{\lambda}_\psi$ differs from $\widehat{\lambda}$ by less than $O_p(n^{-1/2})$ and the asymptotic dependence of $\widehat{\lambda}_\psi$ on ψ is reduced.

To be more explicit, suppose $I_{\lambda\psi}(\psi, \lambda) = 0$ at the true parameter value. Then $\widehat{\lambda} = \widehat{\lambda}_\psi + O_p(n^{-1})$, and it follows that the term $|\partial\widehat{\lambda}/\partial\lambda_\psi|$ that appears in (12.42) equals $1 + O_p(n^{-1})$, in contrast to the value $1 + O_p(n^{-1/2})$ typically obtained. It then seems reasonable to hope that little damage will be done by dropping the Jacobian term from (12.42) and approximating $\ell_{\text{mp}}(\psi)$ by

$$\ell(\psi, \widehat{\lambda}_\psi) - \frac{1}{2} \log |J_{\lambda\lambda}(\psi, \widehat{\lambda}_\psi)|. \quad (12.48)$$

This argument has a serious drawback, because knowledge that a term is $O_p(n^{-1})$ gives no notion of its actual size: $c\psi/n$ is $O(n^{-1})$ both when $c = 100$ and $c = 0.01$, but the numerical values are very different. Asymptotic arguments are valuable heuristics but cannot ensure accuracy in applications. With this in mind, we nevertheless press forward with vigour.

If $I_{\lambda\psi}(\psi, \lambda) = 0$ for all (ψ, λ) , then the parameters λ and ψ are said to be *orthogonal*. Among the consequences of this is that the inverse information matrix is block diagonal, so the maximum likelihood estimators $\widehat{\psi}$ and $\widehat{\lambda}$ are asymptotically independent, and the asymptotic standard error for $\widehat{\psi}$ when λ is unknown is the same as when it is known. Another advantage of is that likelihood maximization may be numerically more stable in the orthogonal parametrization. Below we briefly consider the implications of parameter orthogonality for likelihood modification, first outlining how to obtain parameters orthogonal to a given interest parameter.

Consider a model with log likelihood $\ell^*(\psi, \gamma)$ in terms of the scalar interest parameter ψ and an arbitrary nuisance parameter $\gamma = (\gamma_1, \dots, \gamma_q)^T$. We seek $\lambda = \lambda(\psi, \gamma)$

such that λ is orthogonal to ψ . Writing $\gamma = \gamma(\psi, \lambda)$, we have

$$\ell(\psi, \lambda) = \ell^* \{ \psi, \gamma(\psi, \lambda) \},$$

and differentiation with respect to ψ and λ yields

$$\frac{\partial^2 \ell}{\partial \lambda \partial \psi} = \frac{\partial \gamma^\top}{\partial \lambda} \frac{\partial^2 \ell^*}{\partial \gamma \partial \psi} + \frac{\partial \gamma^\top}{\partial \lambda} \frac{\partial^2 \ell^*}{\partial \gamma \partial \gamma^\top} \frac{\partial \gamma}{\partial \psi} + \frac{\partial^2 \gamma^\top}{\partial \lambda \partial \psi} \frac{\partial \ell^*}{\partial \gamma}.$$

If λ and ψ are to be orthogonal, this expression must have expectation zero. Hence

$$0 = \frac{\partial \gamma^\top}{\partial \lambda} I_{\gamma\psi}^* + \frac{\partial \gamma^\top}{\partial \lambda} I_{\gamma\gamma}^* \frac{\partial \gamma}{\partial \psi},$$

where $I_{\gamma\psi}^*$ and $I_{\gamma\gamma}^*$ are components of the expected information matrix in the non-orthogonal parametrization. Thus provided that the $q \times q$ matrix $\partial \gamma^\top / \partial \lambda$ is invertible and taking for granted the necessary regularity conditions, we see that $\lambda(\psi, \gamma)$ is a solution of the system of q partial differential equations

$$\frac{\partial \gamma}{\partial \psi} = -I_{\gamma\gamma}^{*-1}(\psi, \gamma) I_{\gamma\psi}^*(\psi, \gamma). \quad (12.49)$$

There is latitude in the choice of λ , because if λ and ψ are orthogonal, then any smooth functions of ψ and of λ are orthogonal. Thus λ can in some cases be chosen to have a desirable property such as directness of interpretation.

When the data are a random sample of size n , the expected information equals $I^*(\psi, \gamma) = ni^*(\psi, \gamma)$, and the partial differential equation (12.49) may be expressed using the information matrix $i^*(\psi, \gamma)$ for a single observation.

It is not necessary to find λ in terms of ψ and γ in order to obtain (12.48). To see this, write $\gamma = \gamma(\psi, \lambda)$ and note that

$$\begin{aligned} \frac{\partial \ell(\psi, \lambda)}{\partial \lambda} &= \frac{\partial \gamma^\top}{\partial \lambda} \frac{\partial \ell^*(\psi, \gamma)}{\partial \gamma}, \\ \frac{\partial^2 \ell(\psi, \lambda)}{\partial \lambda \partial \lambda^\top} &= \frac{\partial^2 \gamma^\top}{\partial \lambda \partial \lambda^\top} \frac{\partial \ell^*(\psi, \gamma)}{\partial \gamma} + \frac{\partial \gamma^\top}{\partial \lambda} \frac{\partial^2 \ell^*(\psi, \gamma)}{\partial \gamma \partial \gamma^\top} \frac{\partial \gamma}{\partial \lambda^\top}. \end{aligned}$$

If the maximum likelihood estimates of λ and γ for fixed ψ are $\hat{\lambda}_\psi$ and $\hat{\gamma}_\psi$, then

$$J_{\lambda\lambda}(\psi, \hat{\lambda}_\psi) = \frac{\partial \gamma(\psi, \hat{\lambda}_\psi)^\top}{\partial \lambda} J_{\gamma\gamma}^*(\psi, \hat{\gamma}_\psi) \frac{\partial \gamma(\psi, \hat{\lambda}_\psi)}{\partial \lambda^\top}.$$

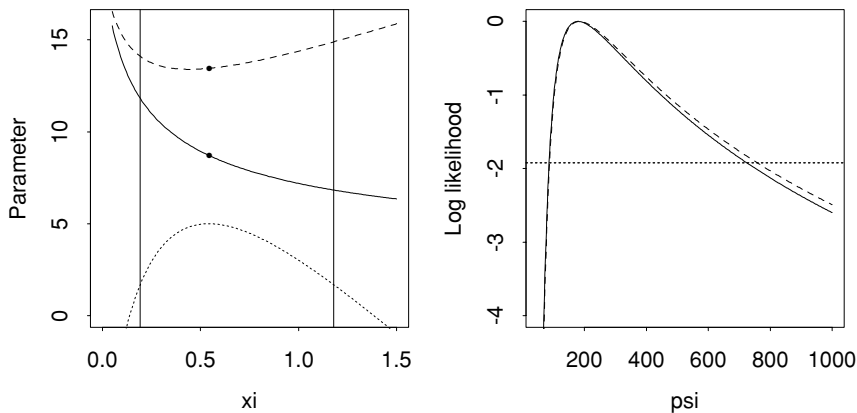
Provided $|\partial \gamma / \partial \lambda^\top| \neq 0$, we have

$$\left| \frac{\partial \gamma(\psi, \hat{\lambda}_\psi)}{\partial \lambda^\top} \right| = \left| \frac{\partial \lambda(\psi, \hat{\gamma}_\psi)}{\partial \gamma^\top} \right|^{-1},$$

and so (12.48) equals

$$\ell^*(\psi, \hat{\gamma}_\psi) - \frac{1}{2} \log |J_{\gamma\gamma}^*(\psi, \hat{\gamma}_\psi)| + \log \left| \frac{\partial \lambda(\psi, \hat{\gamma}_\psi)}{\partial \gamma^\top} \right|, \quad (12.50)$$

which can be computed without writing λ explicitly in terms of ψ and γ .

**Figure 12.11**

Likelihood analysis of Danish fire data. Left: variation of $\hat{\sigma}_\xi$ (solid) and $\hat{\lambda}_\xi$ (dashes) as a function of the shape parameter ξ . The blobs show the maximum likelihood estimates. The dotted line is the profile log likelihood $\ell_p(\xi)$ and the vertical lines mark the limits of a 0.99 confidence interval for ξ . The orthogonal parameter $\hat{\lambda}_\xi$ varies much less than does the non-orthogonal parameter $\hat{\sigma}_\xi$ over the range of ψ considered. Right: profile log likelihood (solid) and modified profile log likelihood (dashes) for 0.99 quantile ψ of generalized Pareto distribution. The horizontal line determines the limits of a 0.95 confidence interval for ψ .

Example 12.25 (Generalized Pareto distribution) The expected information matrix for an observation with distribution function (6.38) is

$$i^*(\xi, \sigma) = \frac{1}{\sigma^2(1+\xi)(1+2\xi)} \begin{pmatrix} 2\sigma^2 & \sigma \\ \sigma & 1+\xi \end{pmatrix}.$$

Hence a parameter $\lambda = \lambda(\xi, \sigma)$ orthogonal to the shape parameter ξ satisfies the partial differential equation

$$\frac{\partial \sigma}{\partial \xi} = -i_{\sigma\sigma}^{*-1}(\xi, \sigma)i_{\sigma\xi}^*(\xi, \sigma) = -\frac{\sigma}{1+\xi}.$$

It is straightforward to check that this implies that $g(\lambda) = \sigma(1+\xi)$ for a suitably smooth function g . With the choice $\lambda = \sigma(1+\xi)$ we find that

$$i(\xi, \lambda) = \frac{1}{\lambda^2(1+\xi)^2(1+2\xi)} \begin{pmatrix} \lambda^2(1+2\xi) & 0 \\ 0 & (1+\xi)^2 \end{pmatrix}.$$

We illustrate numerically the effect of parameter orthogonality using the Danish fire insurance claim data described in Examples 6.31 and 6.34. We apply a threshold $u = 15$ to the claim sizes and fit the generalized Pareto distribution to the resulting 60 exceedances. The left panel of Figure 12.11 shows that $\hat{\sigma}_\xi$ varies more over the range of appreciable likelihood for ξ than does $\hat{\lambda}_\xi = \hat{\sigma}_\xi(1+\xi)$, as our general discussion anticipates.

Now suppose that interest focuses on the $(1-p)$ quantile of the distribution,

$$\psi = \begin{cases} \sigma(p^{-\xi} - 1)/\xi, & \xi \neq 1, \\ -\sigma \log p, & \xi = 0, \end{cases}$$

a return level, to which we seek an orthogonal parameter $\lambda = \lambda(\psi, \xi)$. As

$$i^*(\xi, \psi) = \frac{\partial(\xi, \sigma)^T}{\partial(\xi, \psi)} i^*(\xi, \sigma) \frac{\partial(\xi, \sigma)}{\partial(\xi, \psi)^T} \Big|_{\sigma=\sigma(\xi, \psi)},$$

a tedious calculation shows that $i^*(\xi, \psi)$ may be written as

$$\begin{pmatrix} \psi^2 a(\xi) & -\psi a(\xi)/b(\xi) \\ -\psi a(\xi)/b(\xi) & c(\xi) \end{pmatrix},$$

and so $\lambda = \lambda(\psi, \xi)$ solves the equation $\psi b(\xi) \partial \xi / \partial \psi = 1$. Thus we may take

$$\lambda = g \left\{ \log \psi + \int^{\xi} b(u) du \right\}$$

for any suitably smooth function g . With the choice $g(u) = u$, we have $\partial \lambda / \partial \xi = b(\xi)$, and this may be used in (12.50).

The right panel of Figure 12.11 shows the effect of likelihood modification when setting a confidence interval for the 0.99 quantile of the generalized Pareto model using the 60 exceedances. Application of standard chi-squared asymptotics to the profile log likelihood yields the highly asymmetric 0.95 confidence interval (86.1, 723.8), while the corresponding confidence interval based on the modified profile log likelihood is (87.6, 757.4).

Although modification changes the right-hand limit of the confidence interval appreciably, of overwhelmingly greater concern in applications would be the representativeness of the largest few observations in the sample, on which inferences will hinge. Moreover in practice the quantile also depends on the Poisson rate of exceedance times, and hence must be orthogonalized with respect to two parameters; see Section 6.5.2. ■

Although (12.49) gives a basis for orthogonalizing γ with respect to a scalar ψ , parameters orthogonal to a vector ψ cannot be found in general. For if ψ contains ψ_1 and ψ_2 , say, then λ must simultaneously satisfy both systems of equations

$$\frac{\partial \gamma}{\partial \psi_1} = -I_{\gamma\gamma}^{*-1}(\psi, \gamma) I_{\gamma\psi_1}^*(\psi, \gamma), \quad \frac{\partial \gamma}{\partial \psi_2} = -I_{\gamma\gamma}^{*-1}(\psi, \gamma) I_{\gamma\psi_2}^*(\psi, \gamma),$$

for all γ , ψ_1 and ψ_2 . However there is no guarantee that the compatibility condition $\partial^2 \gamma / \partial \psi_1 \partial \psi_2 = \partial^2 \gamma / \partial \psi_2 \partial \psi_1$ will hold; if not, a simultaneous joint solution does not exist. When a solution exists, it can produce familiar results.

Example 12.26 (Linear exponential family) In a linear exponential family with log likelihood

$$\ell^*(\psi, \gamma) \equiv t_1^T \psi + t_2^T \gamma - \kappa(\psi, \gamma),$$

the parameters $\lambda = \lambda(\psi, \gamma)$ orthogonal to ψ are determined by

$$\frac{\partial \gamma}{\partial \psi^T} = -\kappa_{\gamma\gamma}^{-1}(\psi, \gamma) \kappa_{\gamma\psi}(\psi, \gamma). \quad (12.51)$$

If we reparametrize in terms of ψ and $\lambda = \kappa_{\gamma}(\psi, \gamma) = \partial \kappa(\psi, \gamma) / \partial \gamma$, then in this new parametrization, γ is a function of ψ and λ , and

$$0 = \frac{\partial \lambda^T}{\partial \psi} = \frac{\partial \gamma^T}{\partial \psi} \kappa_{\gamma\gamma}(\psi, \gamma) + \kappa_{\psi\gamma}(\psi, \gamma),$$

so $\lambda = \kappa_{\gamma}(\psi, \gamma)$ is a solution to (12.51). That is, the parameter orthogonal to ψ is the so-called complementary mean parameter $\lambda(\psi, \gamma) = E(T_2; \psi, \gamma)$. By symmetry, $E(T_1; \psi, \gamma)$ is orthogonal to γ .

The normal distribution with mean μ and variance σ^2 has canonical parameter $(\mu/\sigma^2, -1/(2\sigma^2))$. The canonical statistic (Y, Y^2) has expectation $(\mu, \mu^2 + \sigma^2)$, so μ is orthogonal to $-1/(2\sigma^2)$, and hence to σ^2 , while μ/σ^2 is orthogonal to $\mu^2 + \sigma^2$.

Independent Poisson variables Y_1 and Y_2 with means $\exp(\gamma)$ and $\exp(\gamma + \psi)$ have log likelihood

$$\ell^*(\psi, \gamma) \equiv (y_1 + y_2)\gamma + y_2\psi - e^\gamma - e^{\gamma+\psi}.$$

The discussion above suggests that

$$\lambda = E(Y_1 + Y_2) = \exp(\gamma) + \exp(\gamma + \psi) = e^\gamma(1 + e^\psi)$$

is orthogonal to ψ , so $\gamma = \log \lambda - \log(1 + e^\psi)$ and

$$\ell(\psi, \lambda) \equiv y_2\psi - (y_1 + y_2)\log(1 + e^\psi) + (y_1 + y_2)\log \lambda - \lambda.$$

The separation of ψ and λ implies that the profile and modified profile likelihoods for ψ are proportional. They correspond to the conditional likelihood obtained from the density of Y_2 given $Y_1 + Y_2$. ■

Example 12.27 (Restricted likelihood) If $Y \sim N_n(X\beta, \sigma^2\Upsilon^{-1})$, where the parameter of interest ψ appears in the $n \times n$ matrix Υ but not in β , the log likelihood is

$$\ell(\beta, \sigma^2, \psi) \equiv -\frac{n}{2} \log \sigma^2 + \frac{1}{2} \log |\Upsilon| - \frac{1}{2\sigma^2} (y - X\beta)^\top \Upsilon (y - X\beta),$$

differentiation of which yields $\partial \ell / \partial \beta = \sigma^{-2} \Upsilon (y - X\beta)$. It follows that β is orthogonal to both σ^2 and ψ . Now $J_{\beta\beta}(\beta, \sigma^2, \psi) = \sigma^{-2} X^\top \Upsilon X$, so apart from the term $|\partial \hat{\beta}_{\psi, \sigma^2} / \partial \hat{\beta}|^{-1}$, the modified profile log likelihood for ψ and σ^2 equals the marginal log likelihood (12.13). Note that $\hat{\beta}_{\psi, \sigma^2} = (X^\top \Upsilon X)^{-1} X^\top \Upsilon y$ depends on ψ but not on σ^2 .

This argument also applies when the mean of Y is a nonlinear function of β , provided that no parameter appears in both mean and variance. ■

The notion of parameter orthogonality is useful, but the resulting modified likelihoods can be viewed as unsatisfactory, partly because the arbitrariness of the choice of orthogonal parameter results in a lack of uniqueness. A second difficulty is that the partial differential equation and hence its solution will change if there is a minor change to the model, such as the introduction of censoring, so any statistical interpretation of the orthogonal parameter is then compromised. A third is that inferences are typically based on first-order distributional approximations, as mentioned above.

Exercises 12.4

1 Let $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ and let μ be the interest parameter. Show that

$$\ell(\mu, \sigma^2; \hat{\mu}, \hat{\sigma}^2) \equiv -\frac{n}{2} \left\{ \log \sigma^2 + \frac{\hat{\sigma}^2 + (\hat{\mu} - \mu)^2}{\sigma^2} \right\},$$

where $\hat{\mu}$ and $\hat{\sigma}^2$ are the maximum likelihood parameter estimators, and hence find the modified profile likelihood for μ . Compare this with the marginal likelihood based on the t_{n-1} density of $(\bar{Y} - \mu)/(S^2/n)^{1/2}$, where $\bar{Y}$ and S^2 are the unbiased estimators of μ and σ^2 . Discuss.

- 2 Compute (12.43) for (12.15). Hence show that $M(\tau) = |J_{\beta\beta}(\tau, \hat{\beta}_\tau)|^{-1/2}$ for any regression-scale model.
- 3 In Example 12.23, find the asymptotic relative efficiencies of $\hat{\psi}$ and $\hat{\psi}_{\text{mp}}$ when $\psi = 0$.
- 4 Suppose that $y \mapsto (\hat{\psi}, \hat{\lambda}, a)$, that a is ancillary, and that

$$f(\hat{\psi}, \hat{\lambda} \mid a; \psi, \lambda) = f(\hat{\psi} \mid \hat{\lambda}, a; \psi) f(\hat{\lambda} \mid a; \psi, \lambda).$$

By modifying the argument on page 685 show that the first term on the right is proportional to $M(\psi) \exp\{\ell_p(\psi)\}$, with $M(\psi)$ given by (12.42), apart from a relative error of size n^{-1} .

- 5 Independent Poisson variables Y_1 and Y_2 have means $\exp(\gamma)$ and $\exp(\gamma + \psi)$. Find the profile and modified profile log likelihoods for ψ in this parametrization. Comment.
- 6 A Poisson variable Y has mean μ , which is itself a gamma random variable with mean θ and shape parameter ν . Find the marginal density of Y , and show that $\text{var}(Y) = \theta + \theta^2/\nu$ and that ν and θ are orthogonal. Hence show that ν is orthogonal to β for any model in which $\theta = \theta(x^T \beta)$, x being a covariate vector. Is the same true for the model in which $\nu = \theta/\kappa$, so that $\text{var}(Y) = (1 + \kappa)\mu$? Discuss the implications for inference on β when the variance function is unknown.
- 7 Let X and Y be independent exponential variables with means γ^{-1} and $(\gamma\psi)^{-1}$. Show that the parameter $\lambda(\gamma, \psi)$ orthogonal to ψ is the solution to the equation $\partial\gamma/\partial\psi = -\gamma/(2\psi)$, and verify that taking $\lambda = \gamma/\psi^{-1/2}$ yields an orthogonal parametrization. Investigate how this solution changes when X and Y are subject to Type I censoring at c .
- 8 Consider n pairs of independent binomial variables with denominators m_{0j} and m_{1j} and success probabilities

$$\frac{\exp(\lambda_j)}{1 + \exp(\lambda_j)}, \quad \frac{\exp(\lambda_j + \psi)}{1 + \exp(\lambda_j + \psi)}, \quad j = 1, \dots, n.$$

Find a parameter orthogonal to ψ and hence obtain a modified profile likelihood for ψ . How does it compare to that in Example 12.23?

12.5 Bibliographic Notes

R. A. Fisher (1934) suggested that conditioning plays a central role in inference, building on earlier work, and subsequently developed this notion largely through examples; see Fisher (1922, 1925, 1935b, 1956, 1990). Although many of these examples are convincing, a fully satisfactory systematic development remains elusive, owing among other things to the non-existence of exact ancillary statistics in some important models and their non-uniqueness in others. However very substantial progress has been made over the past two decades, initially stemming from Efron and Hinkley (1978) and Barndorff-Nielsen and Cox (1979). The p^* formula was crystallized as being central by Barndorff-Nielsen (1983), building on earlier work going back to Fisher (1934), though a fully general proof is hard to establish; perhaps the most satisfactory is given by Skovgaard (1990). Reid (1995, 2003) gives excellent reviews of the roles in inference of conditioning and asymptotics, while Barndorff-Nielsen and Cox (1994) and Severini (2000) give more extended accounts of modern likelihood asymptotics and many further references.

Conditional and marginal likelihoods were introduced by Bartlett (1937) and are now widely used in applications. Restricted maximum likelihood estimation was first described by Patterson and Thompson (1971), though restricted likelihood itself had been obtained by Hartley and Rao (1967) and has earlier roots. Diggle *et al.* (1994) discuss its use in the context of longitudinal data. Example 12.11 is based on Cruddas *et al.* (1989). Kalbfleisch and Prentice (1973) show the link between marginal and partial likelihoods for the proportional hazards model.

The use of Monte Carlo simulation for conditional testing is described by Besag and Clifford (1989, 1991), and applied extensively in contingency tables by Forster *et al.* (1996) and Smith *et al.* (1996). See also Section 4.2 of Davison and Hinkley (1997).

Saddlepoint approximation was introduced into statistics in the pioneering paper of Daniels (1954), which was largely ignored until interest in small-sample asymptotics revived in the 1970s, focussed particularly by Barndorff-Nielsen and Cox (1979). Reid (1988) surveys statistical aspects of saddlepoint approximation, while Barndorff-Nielsen and Cox (1989) is a standard reference to these and related procedures. Jensen (1995) is a thorough mathematical treatment. Distribution function approximations are described by Daniels (1987), Skovgaard (1987), and Barndorff-Nielsen (1986). The approximation (12.40) was proposed by Skovgaard (1996) and has itself been approximated by Severini (1999). Numerous other approaches have been suggested; see for example the account of likelihoods for component parameters in Fraser (2002) or Chapter 7 of Severini (2000).

Cox and Reid (1987) and its discussion give a systematic treatment of parameter orthogonality and its consequences for conditional inference.

The complexity of many likelihood expressions has led authors such as McCullagh (1987) and Andrews and Stafford (2000) to develop powerful tools for analytic work with or symbolic computation applied to asymptotic expansions.

Brazzale (1999, 2000), and Bellio (1999) describe systematic attempts to implement small-sample likelihood asymptotics for practical use.

12.6 Problems

- 1 Find the Fisher information matrix for Example 12.2, and show that it gives the wrong asymptotic variances for all the parameters.
- 2 Does the argument of Example 12.5 apply to the location-scale model with any error distribution?
- 3 Show that a missingness indicator (Section 5.5.1) is not generally an ancillary statistic, and discuss the implications for inference on θ .
- 4 A normal random variable Y has mean θ and variance determined by Table 12.5, where k_1 and k_2 are chosen so that the values of π lie in the unit interval. Show that A_1 is exactly ancillary but is not informative about the precision of Y , while A_2 is not exactly ancillary but is indicative of the precision of Y .
Discuss briefly the merits of exact and approximate conditional inference here. (Lloyd, 1992)

Table 12.5 Conditional distributions of normal variables.

var(Y)	100	100	1	1
π	$\frac{1}{2}(1 - k_1\theta)$	$\frac{1}{2}(1 + k_2\theta)$	$\frac{1}{2}(1 + k_1\theta)$	$\frac{1}{2}(1 - k_2\theta)$
A_1	1	0	1	0
A_2	0	0	1	1

- 5 Show that if Y has the Cauchy density

$$f(y; \mu, \sigma) = \frac{\sigma}{\pi \{\sigma^2 + (y - \mu)^2\}}, \quad -\infty < y, \mu < \infty, \quad \sigma > 0,$$

then $1/Y$ has density $f(y; \mu', \sigma')$, where $\mu' = \mu/(\mu^2 + \sigma^2)$ and $\sigma' = \sigma/(\mu^2 + \sigma^2)$. Deduce that a random sample $Y_1, \dots, Y_n$ of Cauchy variables yields two distinct sets of maximal ancillary statistics. Discuss inference for μ, σ , and for μ/σ . (McCullagh, 1992)

- 6 In Example 12.8, show that A_1 and A_2 are not jointly ancillary for θ .

Conditioning on an ancillary statistic is intended to divide the sample space into relevant subsets according to their information content, so one basis for choice among competing ancillaries is to take that whose the conditional Fisher information has largest variance. Show that $E\{i_1(\theta)\} = E\{i_2(\theta)\}$, but that $\text{var}\{i_1(\theta)\} > \text{var}\{i_2(\theta)\}$ for $0 < \theta < 1$, and deduce that A_1 is preferable.

Discuss critically this idea. (Cox, 1971)

- 7 Consider two independent exponential random samples, $Y_1, \dots, Y_n$ having rate $\psi > 0$ and $X_1, \dots, X_n$ having rate λ/ψ , where $\lambda > 0$.

(a) Suppose it is required to find an approximate ancillary for ψ when $\lambda = 1$. Find the likelihood ratio statistic for testing $\lambda = 1$ against the alternative putting no restriction on λ , and show that it is a function only of $\bar{X}\bar{Y}$. Hence give an exact ancillary statistic for ψ .

(b) Give explicit expressions for the p^* formula (12.7) and for $r^*(\psi)$ when $\lambda = 1$. Investigate the numerical accuracy of (12.8) when $n = 1$.

- 8 A Poisson process of rate $\lambda e^{\psi t}$ is observed on the interval $[0, 1]$, over which events occur at times $0 < t_1 < \dots < t_n < 1$. Show that the likelihood is

$$\exp \left\{ -\lambda \int_0^1 e^{\psi u} du \right\} \prod_{j=1}^n \lambda e^{\psi t_j},$$

and deduce that inference for ψ may be based on the conditional density

$$n! \prod_{j=1}^n \frac{e^{\psi t_j}}{\int_0^1 e^{\psi u} du}$$

of the times of events $T_1 < \dots < T_N$ given that $N = n$. Show that this is the joint density of the order statistics of a random sample of n variables with density $e^{\psi t} / \int_0^1 e^{\psi u} du$ on $(0, 1)$, and derive a conditional test of the hypothesis $\psi = 0$ against $\psi > 0$, giving the null mean and variance of your test statistic.

When $\psi = 0$, how might you test the hypothesis that the T_j are clustered relative to a Poisson process?

(Cox and Lewis, 1966, pp. 45–51)

- 9 Under the model of Example 1.4, the number of deaths due to lung cancer in the (i, j) cell of Table 1.4, Y_{ij} , is a Poisson variable with mean expressible as

$$x_{ij} g(t_i, \phi) (1 + \psi_1 d_j^{\psi_2}),$$

where the notation reflects our interest in the effect of smoking.

Show that the marginal density of $M_i = Y_{i1} + \cdots + Y_{ic}$ is Poisson with mean

$$\lambda_i = g(t_i; \phi) \sum_{j=1}^c x_{ij}(1 + \psi_1 d_j^{\psi_2}), \quad i = 1, \dots, r,$$

while the conditional distribution of $Y_{i1}, \dots, Y_{ic}$ given $M_i = m_i$ is multinomial with denominator $m_i = y_{i1} + \cdots + y_{ic}$ and probabilities

$$\pi_{ij} = \frac{x_{ij}(1 + \psi_1 d_j^{\psi_2})}{\sum_{k=1}^c x_{ik}(1 + \psi_1 d_k^{\psi_2})}, \quad j = 1, \dots, c.$$

Outline how this computation may be used as a basis for inference on ψ , and in particular how evidence for $\psi_2 = 1$ may be assessed. Do the usual likelihood asymptotics apply when testing the hypothesis that $\psi_1 = 0$, regardless of ψ_2 ?

- 10 In an exponential family density

$$f(t_1, t_2; \psi, \lambda) = \exp \{t_1^T \psi + t_2^T \lambda - \kappa(\psi, \lambda) + c(t_1, t_2)\},$$

show that the conditional distribution of T_1 given T_2 is unchanged if λ is randomly taken from a density $g(\lambda)$.

Independent pairs of observations $(x_1, y_1), \dots, (x_n, y_n)$ are supposed to have independent Poisson distributions with means $(\mu_j, \beta\mu_j)$, for $j = 1, \dots, n$. Does your inference for β

depend on the knowledge that $\mu_1, \dots, \mu_n \stackrel{\text{iid}}{\sim} g$?

If the density $g(\mu) = g(\mu; \gamma)$ is known up to the value of a parameter γ , say, suggest how to retrieve any information on ψ in the marginal density of the y_j .

- 11 Independent pairs of binary observations $(R_{01}, R_{11}), \dots, (R_{0n}, R_{1n})$ have success probabilities $(e^{\lambda_j}/(1 + e^{\lambda_j}), e^{\psi + \lambda_j}/(1 + e^{\psi + \lambda_j}))$, for $j = 1, \dots, n$.

(a) Show that the maximum likelihood estimator of ψ based on the conditional likelihood is $\hat{\psi}_c = \log(R^{01}/R^{10})$, where R^{01} and R^{10} are respectively the numbers of (0,1) and (1,0) pairs. Does $\hat{\psi}_c$ tend to ψ as $n \rightarrow \infty$?

(b) Write down the unconditional likelihood for ψ and λ , and show that the likelihood equations are equivalent to

$$r_{0j} + r_{1j} = \frac{e^{\hat{\lambda}_j}}{1 + e^{\hat{\lambda}_j}} + \frac{e^{\hat{\lambda}_j + \hat{\psi}}}{1 + e^{\hat{\lambda}_j + \hat{\psi}}}, \quad j = 1, \dots, n, \quad (12.52)$$

$$\sum_{j=1}^n r_{1j} = \sum_{j=1}^n \frac{e^{\hat{\lambda}_j + \hat{\psi}}}{1 + e^{\hat{\lambda}_j + \hat{\psi}}}.$$

(i) Show that the maximum likelihood estimator of λ_j is ∞ if $r_{0j} = r_{1j} = 1$ and $-\infty$ if $r_{0j} = r_{1j} = 0$; such pairs are not informative. (ii) Use (12.52) to show that $\hat{\lambda}_j = -\hat{\psi}/2$ for those pairs for which $r_{0j} + r_{1j} = 1$. (iii) Hence deduce that the unconditional maximum likelihood estimator of ψ is $\hat{\psi}_u = 2 \log(R^{01}/R^{10})$. What is the implication for unconditional estimation of ψ ?

- 12 Consider two independent Poisson random samples $X_1, \dots, X_n$ and $Y_1, \dots, Y_n$, the first having mean λ and the second having mean $\lambda\psi$, where $\lambda, \psi > 0$.

(a) Show that $(T_1, T_2) = (X_1 + \cdots + X_n, Y_1 + \cdots + Y_n)$ is minimal sufficient, and for any fixed value of ψ establish that λ may be eliminated by conditioning on $T_\psi = T_1 + \psi T_2$.

(b) Let $(t_{1,\text{obs}}, t_{2,\text{obs}})$ denote the observed value of (T_1, T_2) . Sketch the sample space for (T_1, T_2) , and consider how the relevant subset

$$\{(t_1, t_2) : t_1 + \psi t_2 = t_{1,\text{obs}} + \psi t_{2,\text{obs}}\}$$

varies with ψ . Hence explain how an exact significance level for a test of $\psi = \psi_0$ against the alternative $\psi > \psi_0$ will depend on ψ_0 . Do you find this satisfactory?

- 13 Adapt the argument giving inference for the regression-scale model to the location-scale model $Y = \mu + e^\tau \varepsilon$, and outline how to make small-sample inferences for μ and for τ .

What happens if ψ is rational? What if ψ is irrational?

Compare the resulting confidence intervals for μ with the the posterior credible intervals found by Bayesian inference using prior density $\pi(\mu, \sigma) \propto \sigma^{-1}$ and distribution function approximation (11.31). Discuss.

- 14 (a) Consider a location-scale model in which $y = \eta + \sigma\varepsilon$, where ε has a known density g . Find parameters orthogonal to η and to σ , and give conditions under which η and σ are themselves orthogonal.
 (b) Consider a regression model $y = x^\top \beta + \sigma\varepsilon$, where ε again has density g . Find a parameter orthogonal to the first component of β , and compare your result with the discussion following (8.8).
- 15 Independent exponential variables $Y_1, \dots, Y_n$ have means $E(Y_j) = \lambda e^{x_j \psi}$, where $\sum x_j = 0$. Show that λ and ψ are orthogonal parameters.
 (a) Show that $\hat{\lambda}_\psi = n^{-1} \sum e^{x_j \psi}$, and deduce that the likelihood ratio statistic for testing $\psi = 0$ can be written as $\mathcal{W}p(0) = 2n \log(\hat{\lambda}_0/\hat{\lambda}_\psi)$.
 (b) Let $\gamma = \log \lambda$. By writing the model in linear regression form, show that $\partial \hat{\gamma} / \partial \hat{\gamma}_\psi = 1$, and deduce that $\partial \hat{\lambda} / \partial \hat{\lambda}_\psi = \hat{\lambda} / \hat{\lambda}_\psi$. Hence find the modified profile likelihood both with and without this term, and compare the resulting likelihood ratio statistics with $\mathcal{W}p(0)$. (Cox and Reid, 1987)
- 16 The Michaelis–Menton model of nonlinear regression is usually specified as

$$Y_j = \frac{\beta_0 x_j}{\beta_1 + x_j} + \varepsilon_j, \quad \varepsilon_1, \dots, \varepsilon_n \stackrel{\text{iid}}{\sim} N(0, \sigma^2);$$

we assume that σ^2 is known. Show that the log likelihood is

$$\ell^*(\beta_0, \beta_1) = -\frac{1}{2\sigma^2} \sum_{j=1}^n \{y_j - \beta_0 x_j / (\beta_1 + x_j)\}^2,$$

and find the expected information matrix.

- (a) Show that the parameter $\lambda = \lambda(\beta_1, \beta_0)$ orthogonal to β_1 is determined by

$$g(\lambda) = \beta_0^2 \sum_{j=1}^n \frac{x_j^2}{(\beta_1 + x_j)^2}$$

for an appropriate smooth function g . Choose g suitably, and write the log likelihood explicitly in terms of λ and β_1 .

- (b) Show that the parameter $\lambda = \lambda(\beta_1, \beta_0)$ orthogonal to β_0 is determined by

$$\beta_0 \sum_{j=1}^n \frac{x_j^2}{(\beta_1 + x_j)^4} \frac{\partial \beta_1}{\partial \beta_0} = \sum_{j=1}^n \frac{x_j^2}{(\beta_1 + x_j)^3}$$

and check that its solution is

$$g_1(\lambda) = \beta_0^3 \sum_{j=1}^n \frac{x_j^2}{(\beta_1 + x_j)^3}.$$

Can the log likelihood be expressed explicitly in terms of λ and β_0 ?

- (c) The orthogonal parametrizations above depend on the design points x_j . Do you find this satisfactory?
 (Hills, 1987)